Hodoscope Paddle Efficiencies

Comparison: RS67 vs RS57-70 Final

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Input Data

Input files used:

- `effies_rs57.xlsx`
- `effies_rs59.xlsx`
- `effies_rs62.xlsx`
- `effies_rs67.xlsx`
- `effies_rs70.xlsx`

Note

The current study is only focused on runs 2 and 3 only. We will do the same study for runs 5 and 6 later as a separate study.

Analysis Procedure

- **Average Efficiency:** Calculated the arithmetic mean of efficiency across the runs.
- **Propagated Error:** Errors were combined in quadrature:

$$\sigma_{prop} = \frac{1}{N}\sqrt{\sum_{i=1}^{N}\sigma_i^2}$$

- **Standard Deviation (σ_{sample}):** Calculated the sample standard deviation of the efficiency distribution to determine the spread:

$$\sigma_{sample} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(\epsilon_i - \bar{\epsilon})^2}$$

- **Error Bar Substitution:** If the standard deviation is larger than the maximum propagated error, both lower and upper error bars are replaced by it:

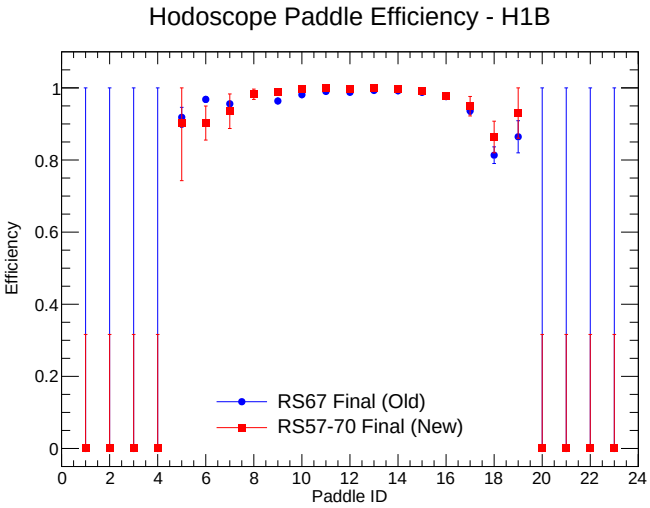
$$\text{If } \sigma_{sample} > \max(\sigma_{prop}^{up}, \sigma_{prop}^{down}) \implies \sigma^{up,down} = \sigma_{sample}$$

- **Physical Limit Capping:** To respect physical boundaries, if the average efficiency plus

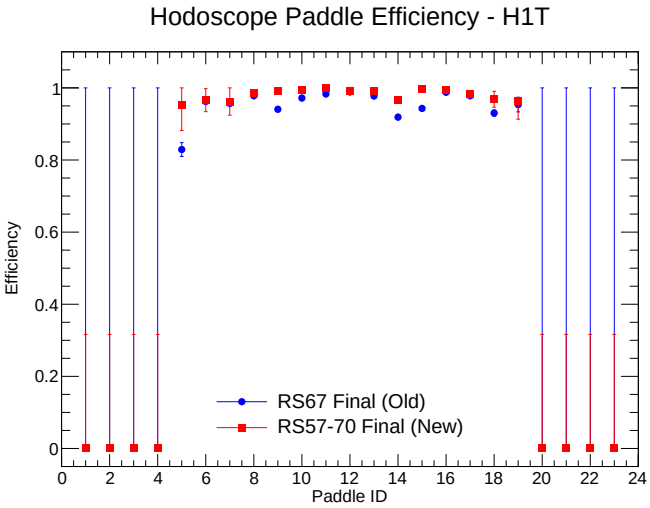
Paddles with Upper Error Bar Capped at 1.0 Bound

Plane	Paddle ID	Original σ_{up}	Capped σ_{up}
H1B	5	0.1592	0.0981
H1B	19	0.0707	0.0690
H1T	5	0.0704	0.0479
H1T	7	0.0378	0.0378
H2B	2	0.5164	0.4000
H2B	8	0.0006	0.0005
H2T	8	0.0004	0.0004
H2T	9	0.0004	0.0004
H3B	4	0.0086	0.0075
H3B	16	0.0318	0.0182
H3T	8	0.0011	0.0004

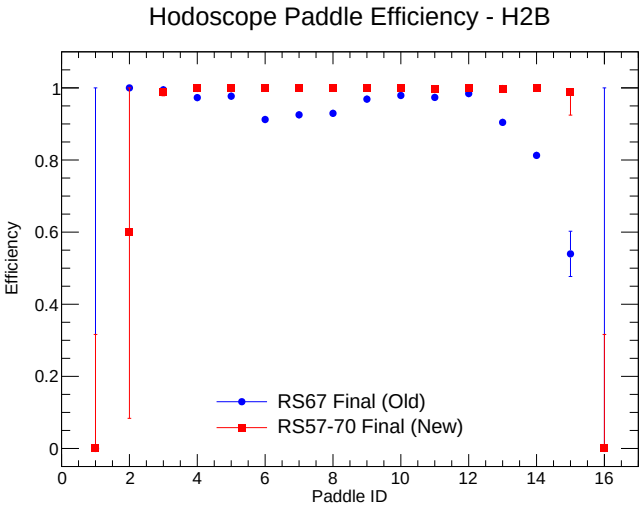
Hodoscope Efficiency: H1B



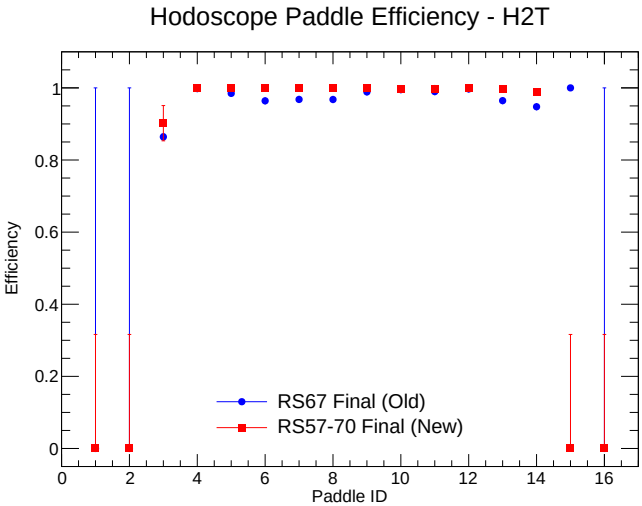
Hodoscope Efficiency: H1T



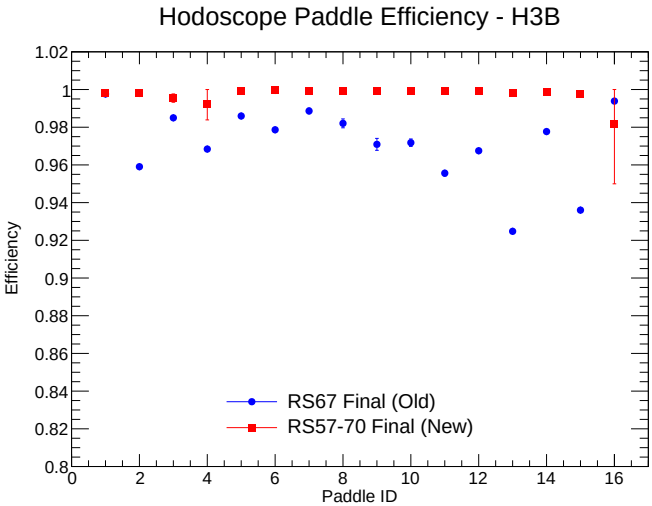
Hodoscope Efficiency: H2B



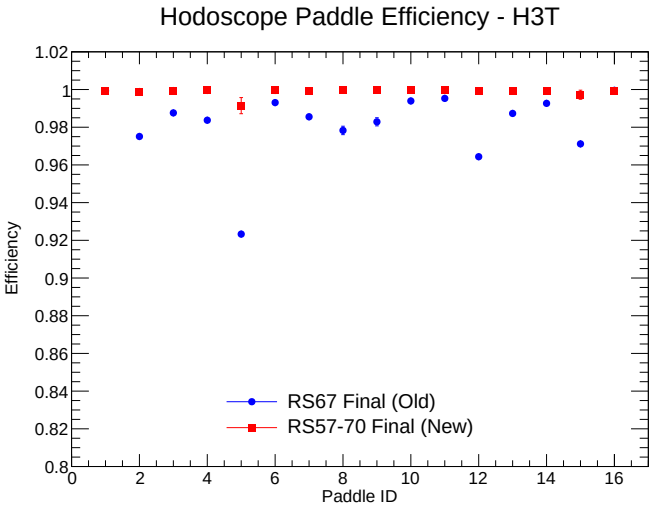
Hodoscope Efficiency: H2T



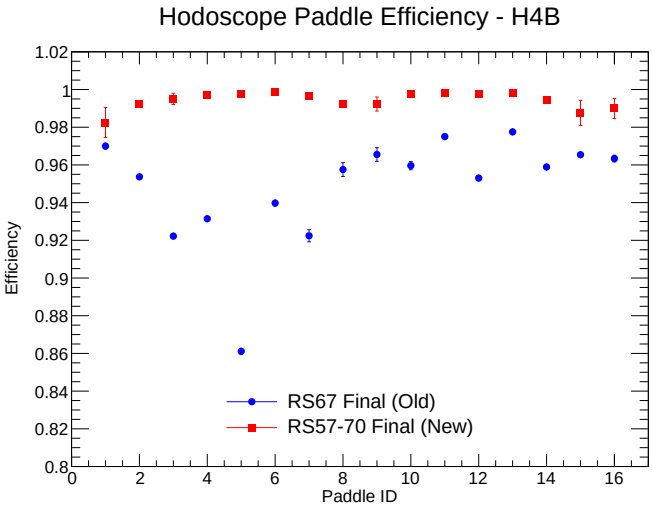
Hodoscope Efficiency: H3B



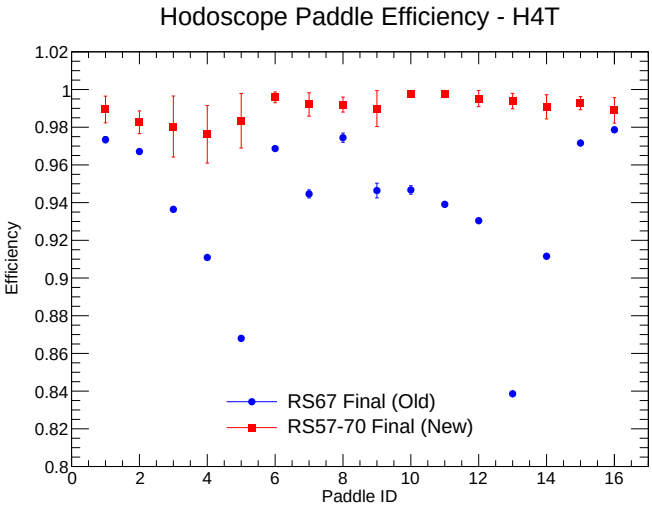
Hodoscope Efficiency: H3T



Hodoscope Efficiency: H4B



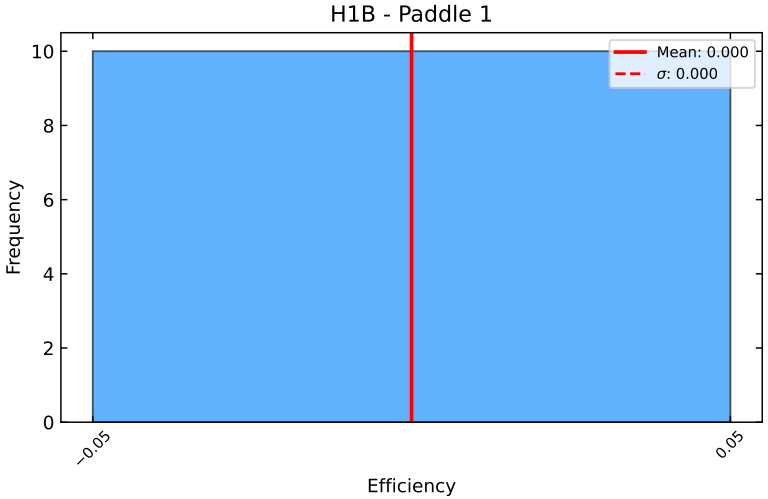
Hodoscope Efficiency: H4T



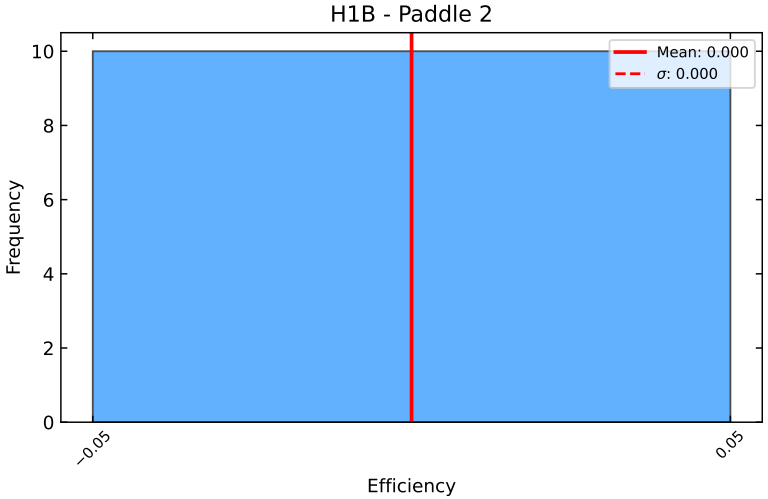
Appendix

Hodoscope Efficiency Distributions with Mean and RMS

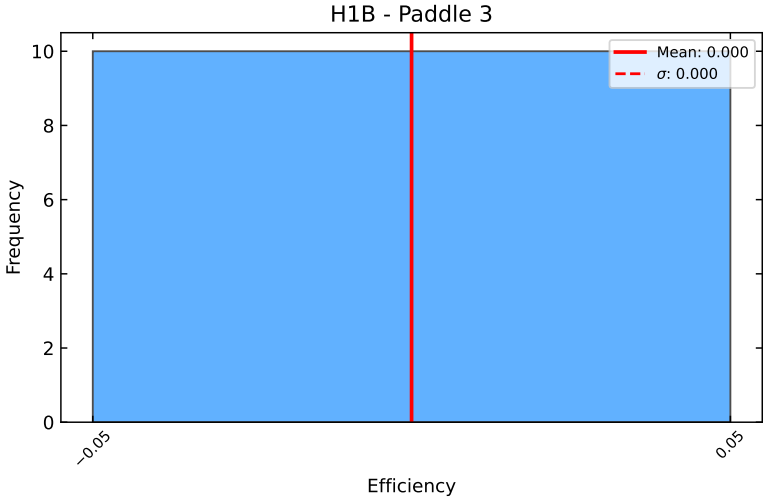
Efficiency Distribution: H1B - Paddle 1



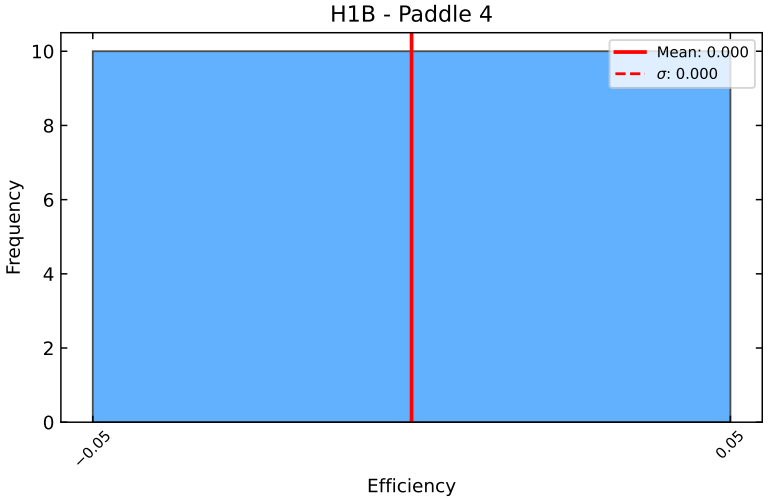
Efficiency Distribution: H1B - Paddle 2



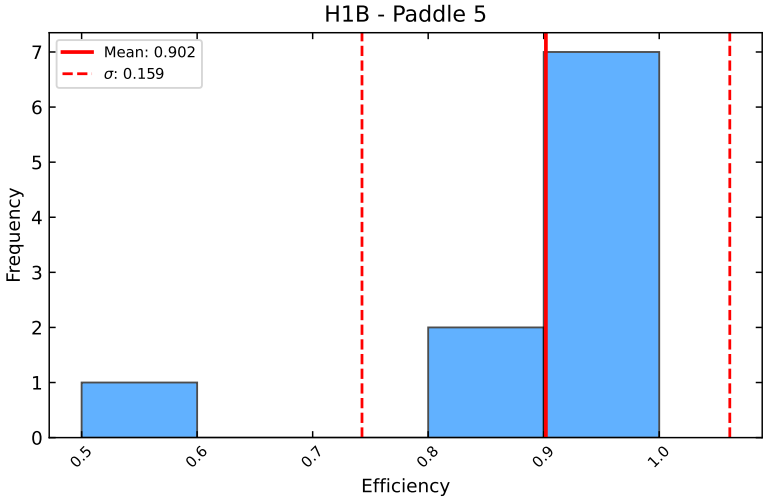
Efficiency Distribution: H1B - Paddle 3



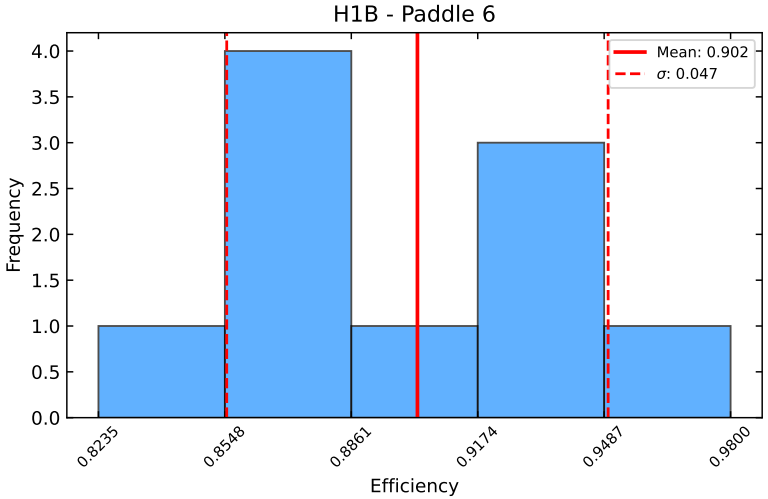
Efficiency Distribution: H1B - Paddle 4



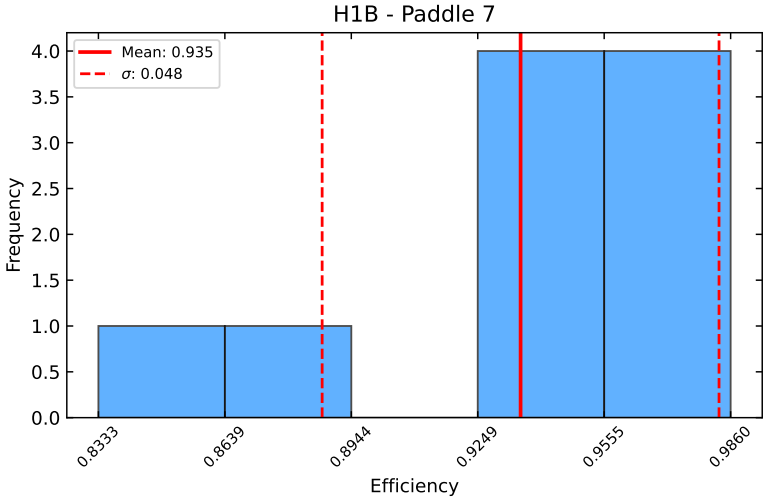
Efficiency Distribution: H1B - Paddle 5



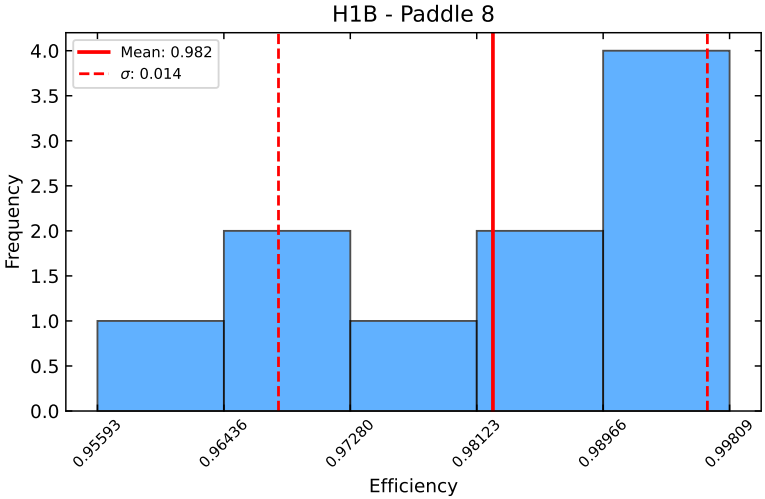
Efficiency Distribution: H1B - Paddle 6



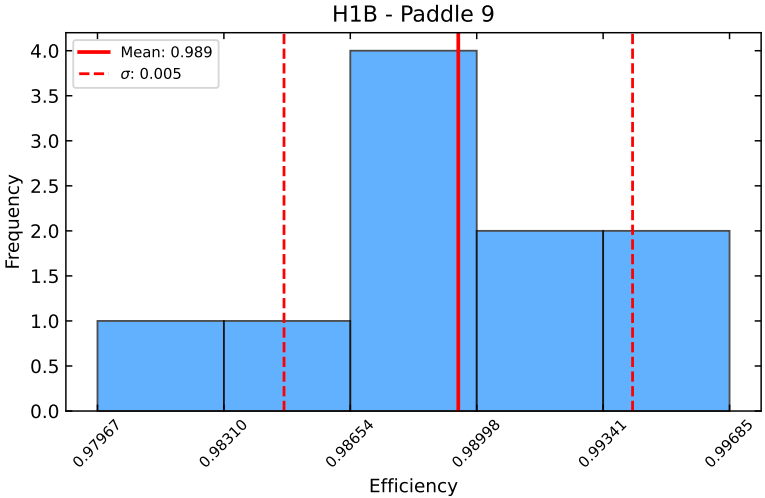
Efficiency Distribution: H1B - Paddle 7



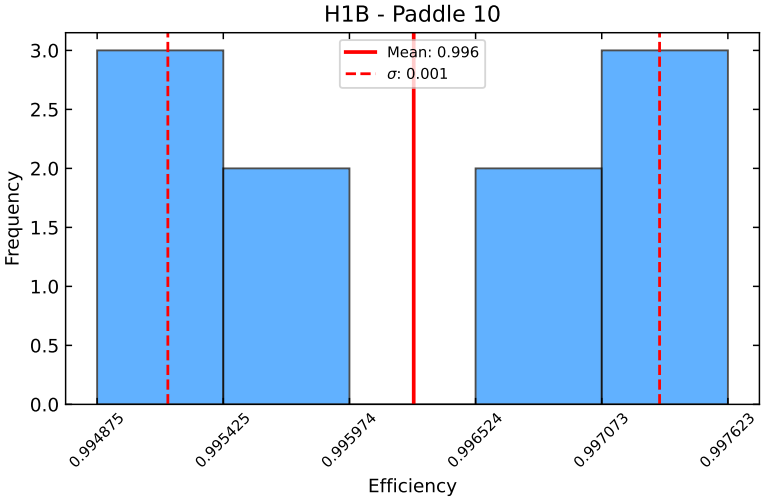
Efficiency Distribution: H1B - Paddle 8



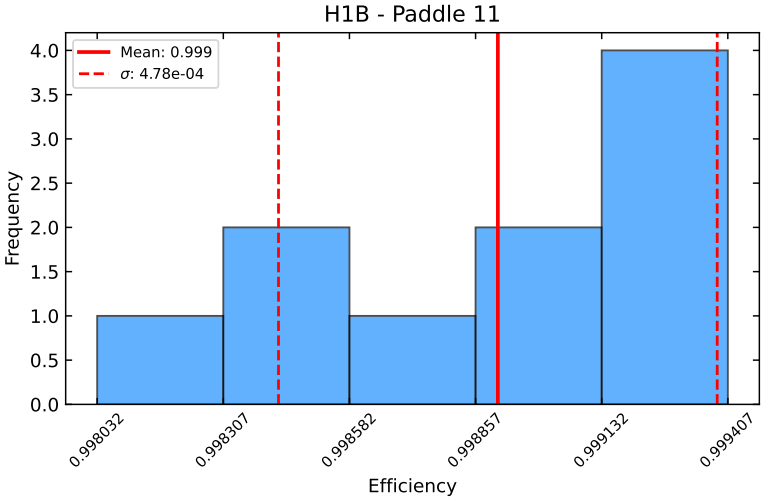
Efficiency Distribution: H1B - Paddle 9



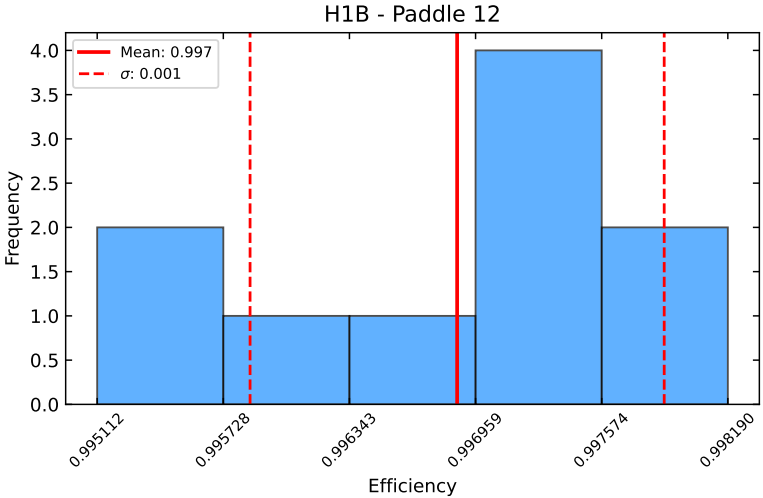
Efficiency Distribution: H1B - Paddle 10



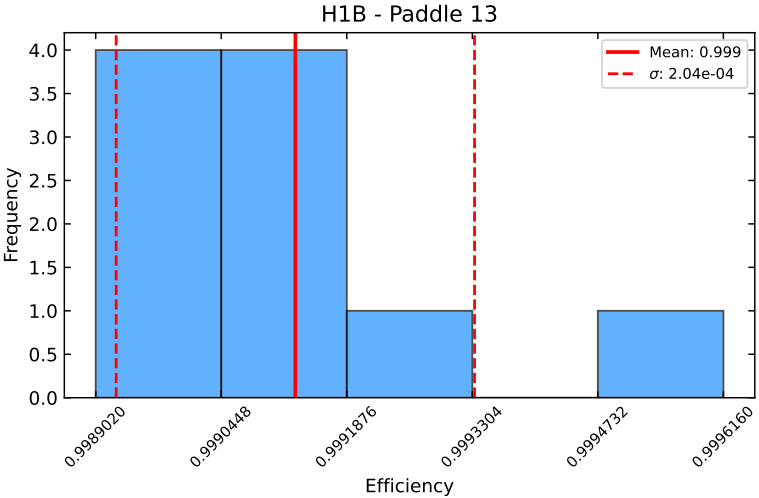
Efficiency Distribution: H1B - Paddle 11



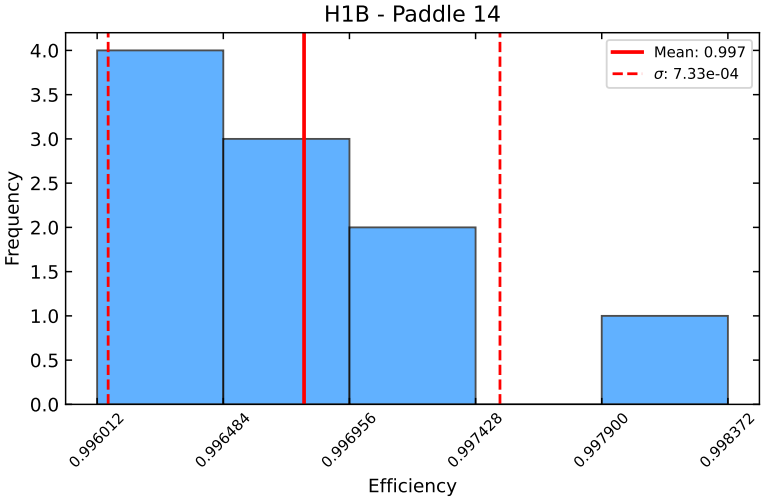
Efficiency Distribution: H1B - Paddle 12



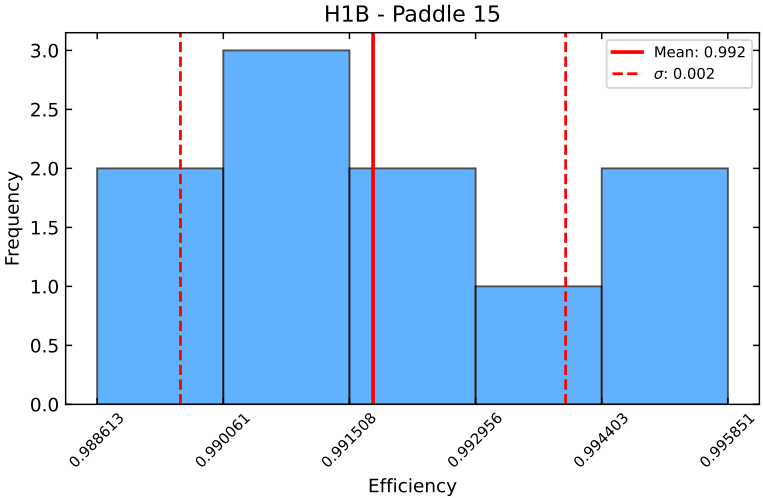
Efficiency Distribution: H1B - Paddle 13



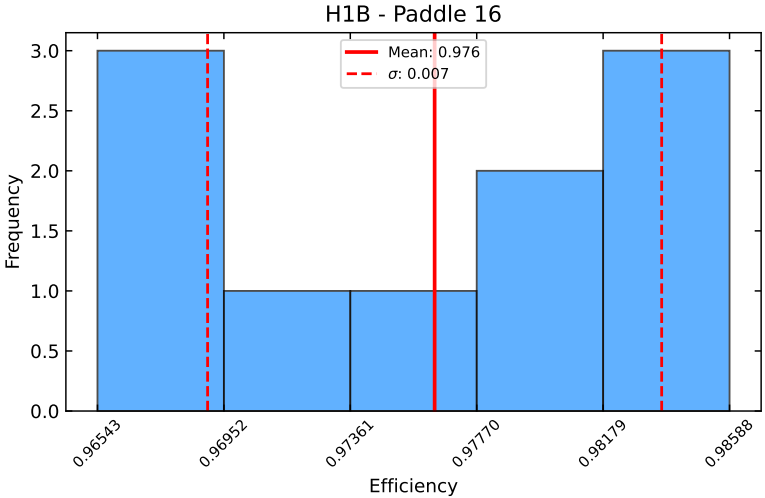
Efficiency Distribution: H1B - Paddle 14



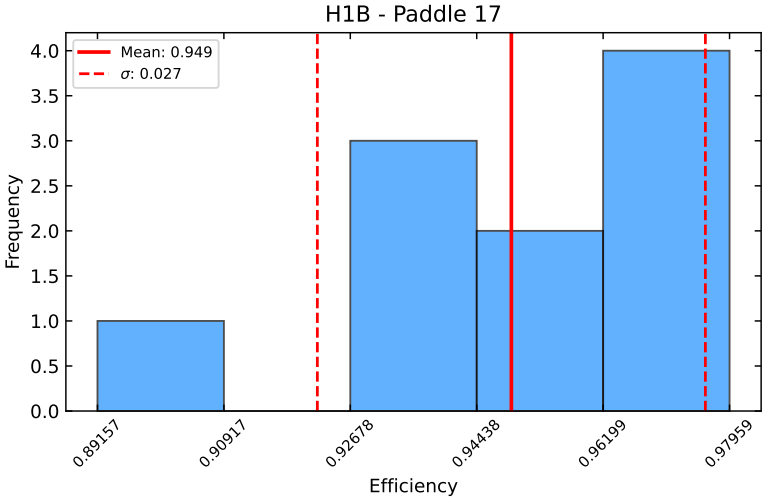
Efficiency Distribution: H1B - Paddle 15



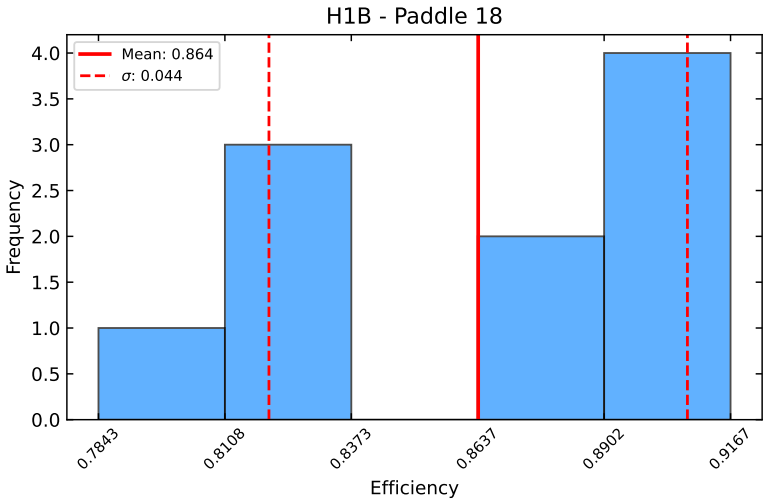
Efficiency Distribution: H1B - Paddle 16



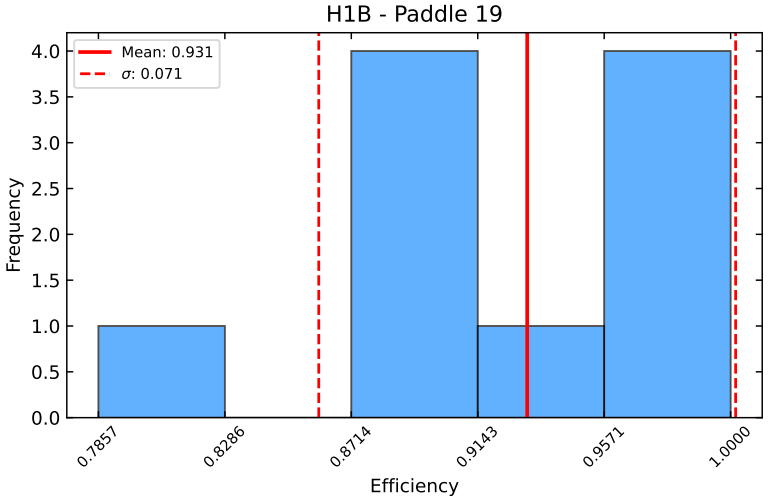
Efficiency Distribution: H1B - Paddle 17



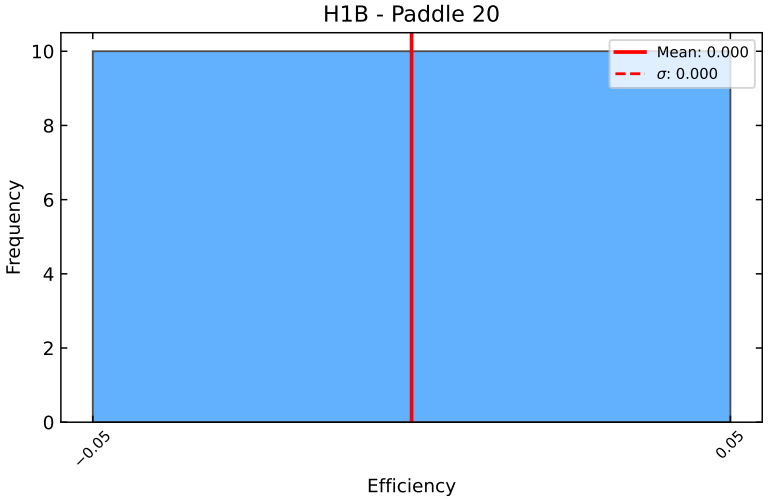
Efficiency Distribution: H1B - Paddle 18



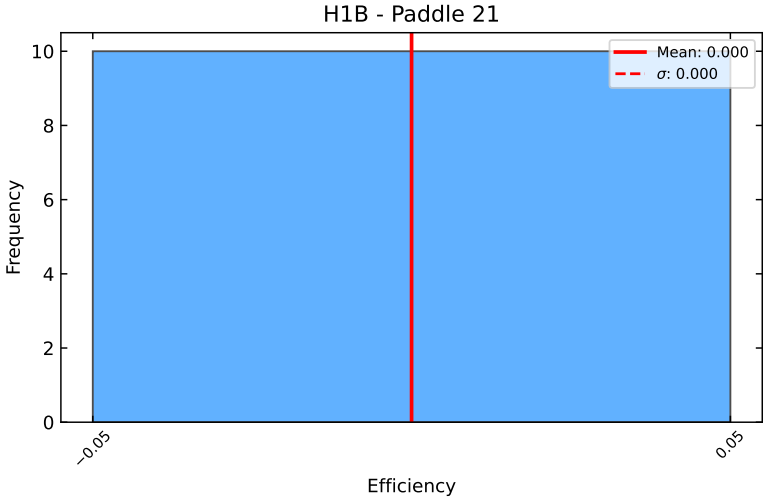
Efficiency Distribution: H1B - Paddle 19



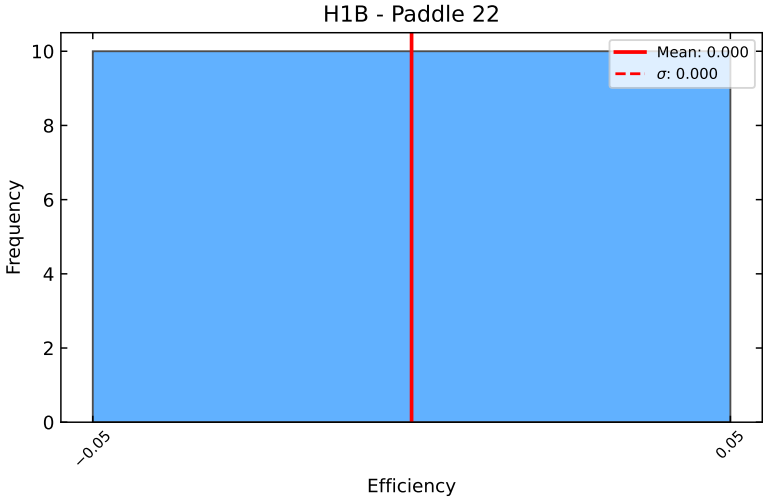
Efficiency Distribution: H1B - Paddle 20



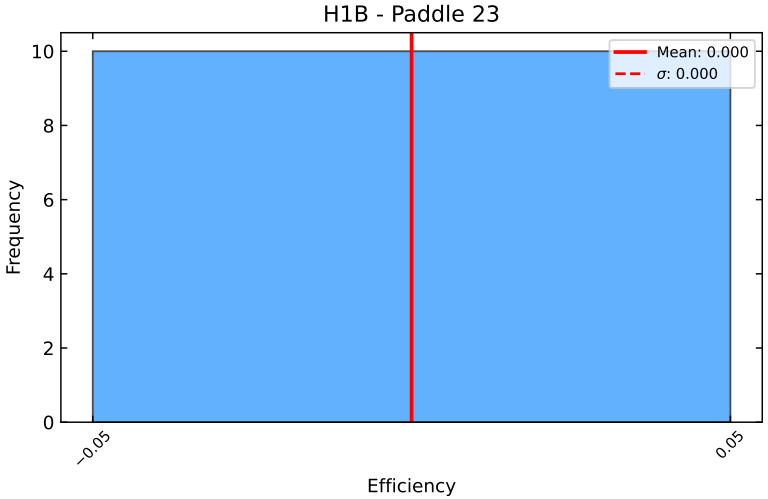
Efficiency Distribution: H1B - Paddle 21



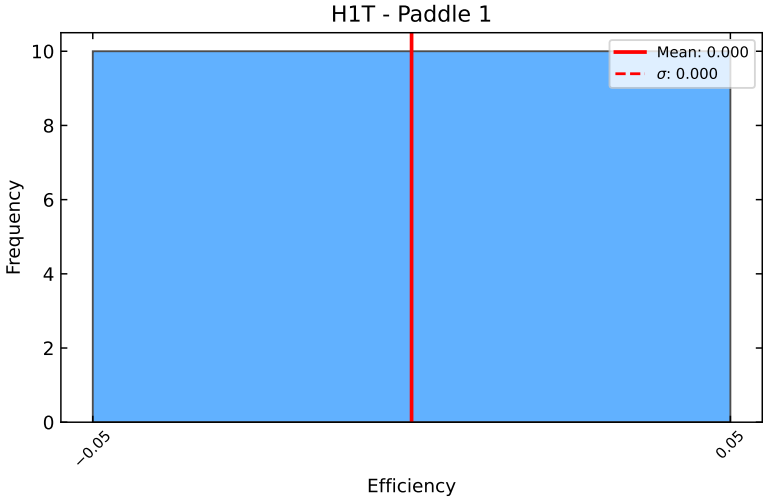
Efficiency Distribution: H1B - Paddle 22



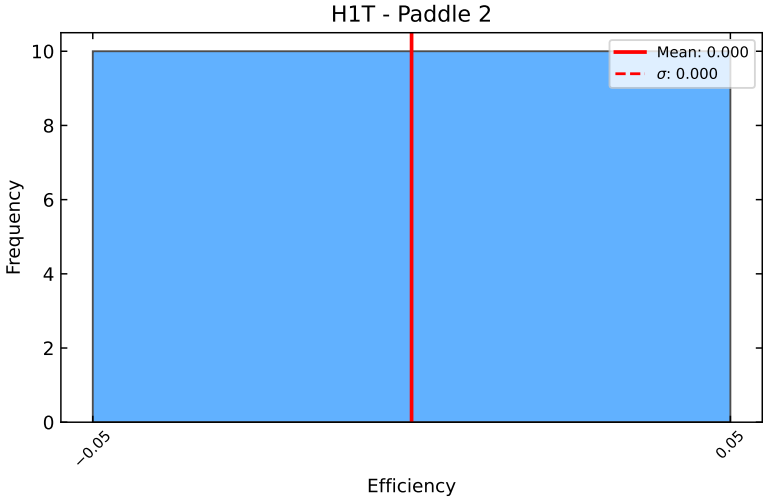
Efficiency Distribution: H1B - Paddle 23



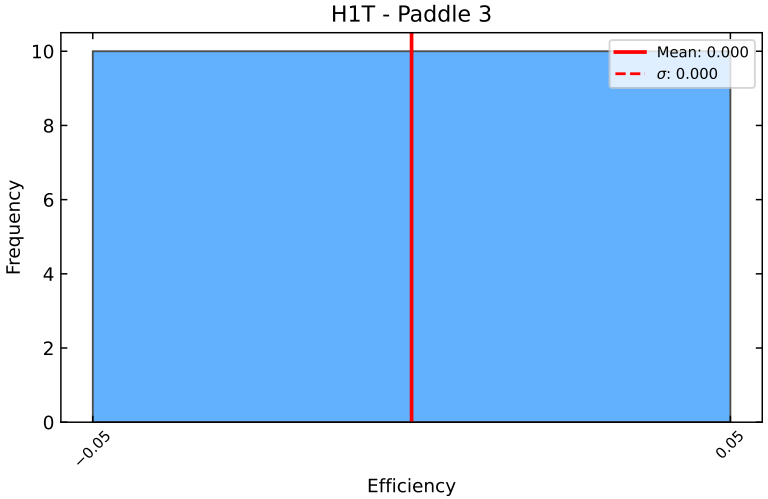
Efficiency Distribution: H1T - Paddle 1



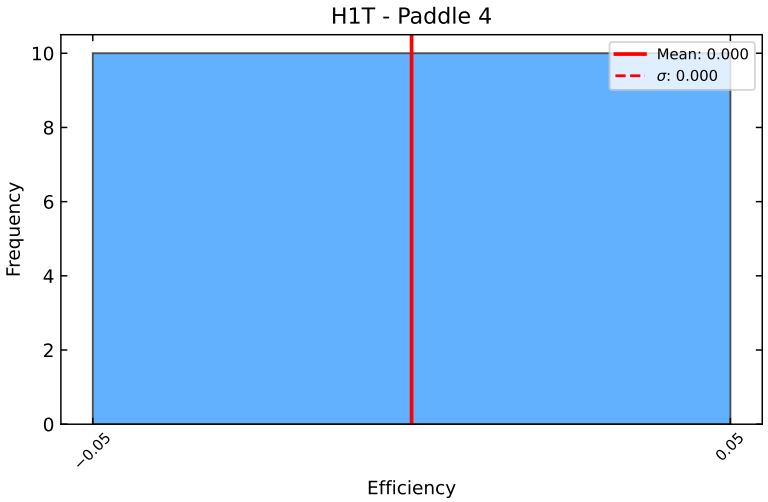
Efficiency Distribution: H1T - Paddle 2



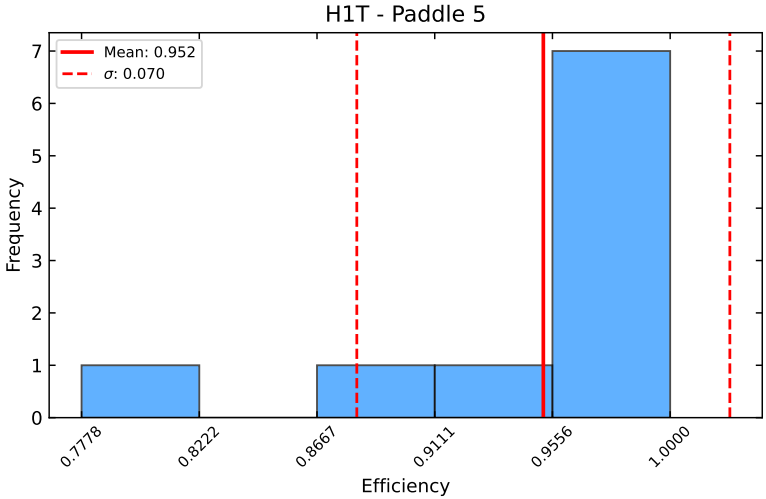
Efficiency Distribution: H1T - Paddle 3



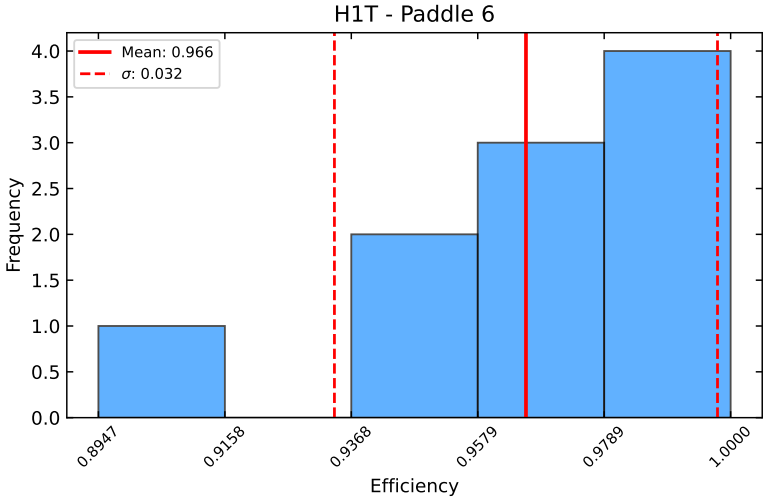
Efficiency Distribution: H1T - Paddle 4



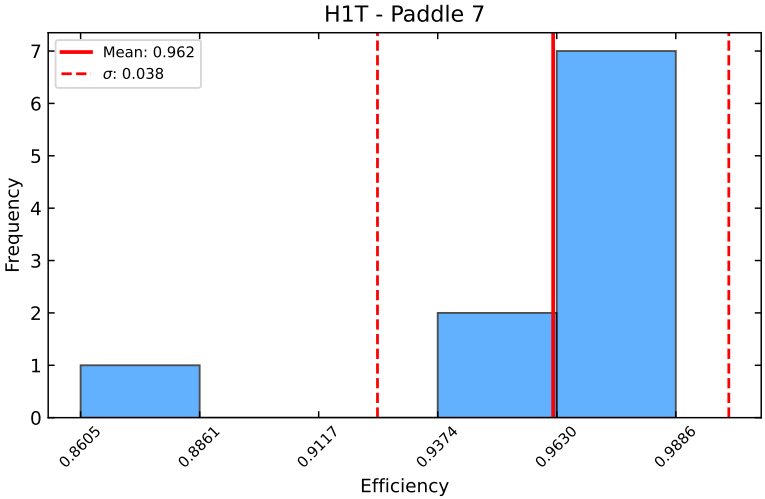
Efficiency Distribution: H1T - Paddle 5



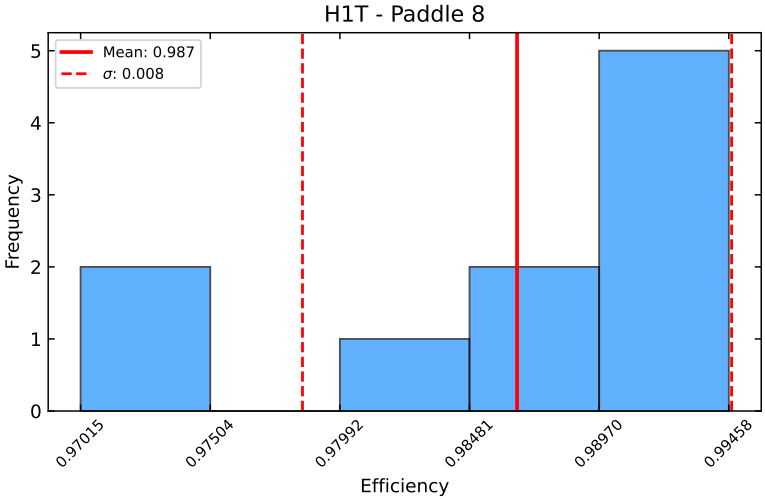
Efficiency Distribution: H1T - Paddle 6



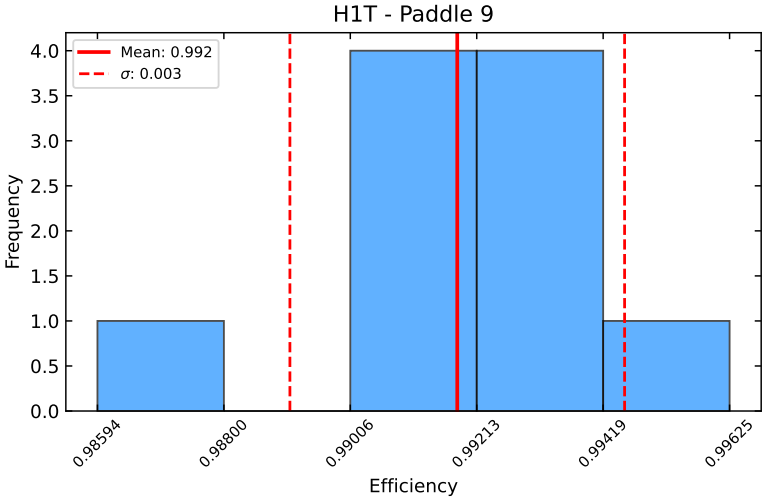
Efficiency Distribution: H1T - Paddle 7



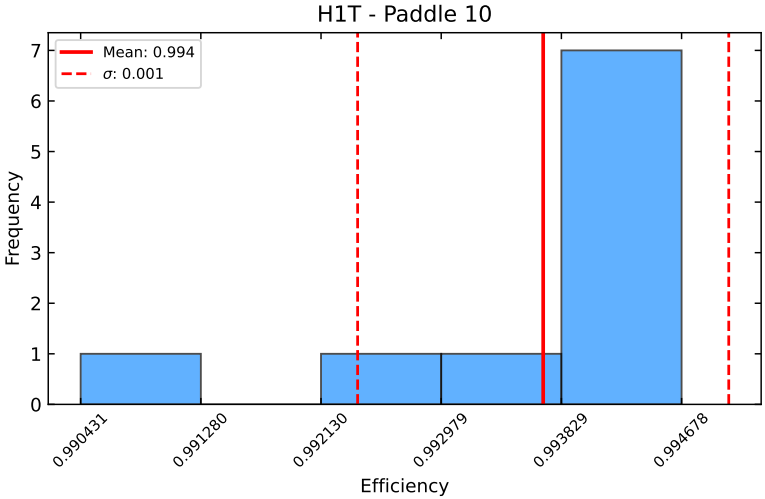
Efficiency Distribution: H1T - Paddle 8



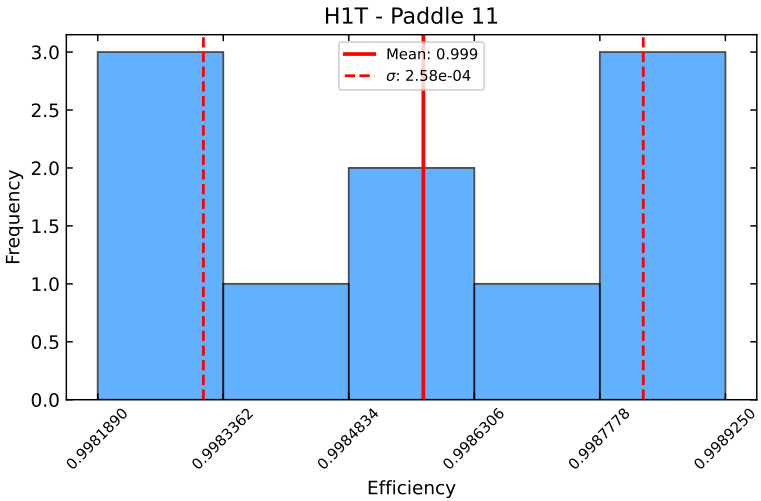
Efficiency Distribution: H1T - Paddle 9



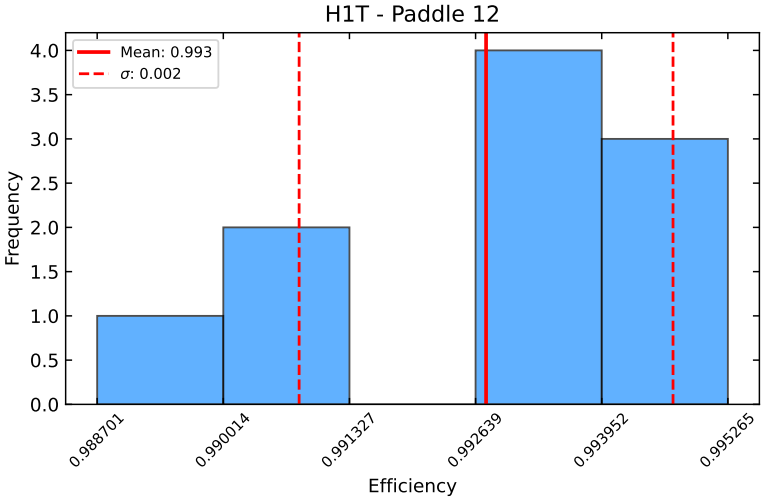
Efficiency Distribution: H1T - Paddle 10



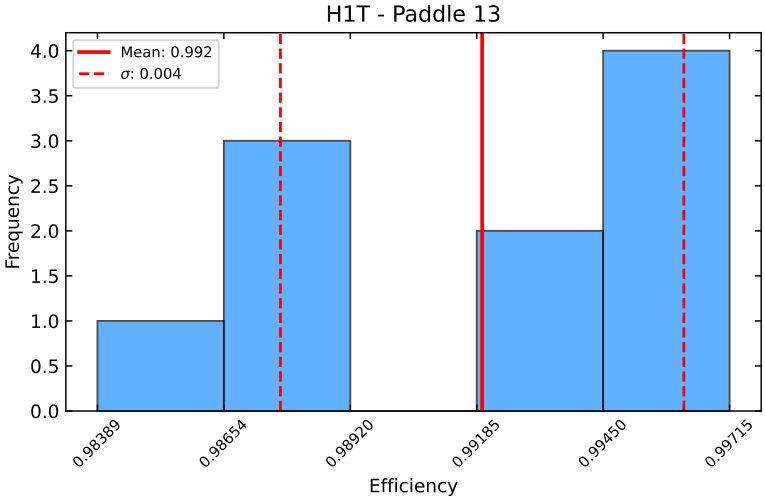
Efficiency Distribution: H1T - Paddle 11



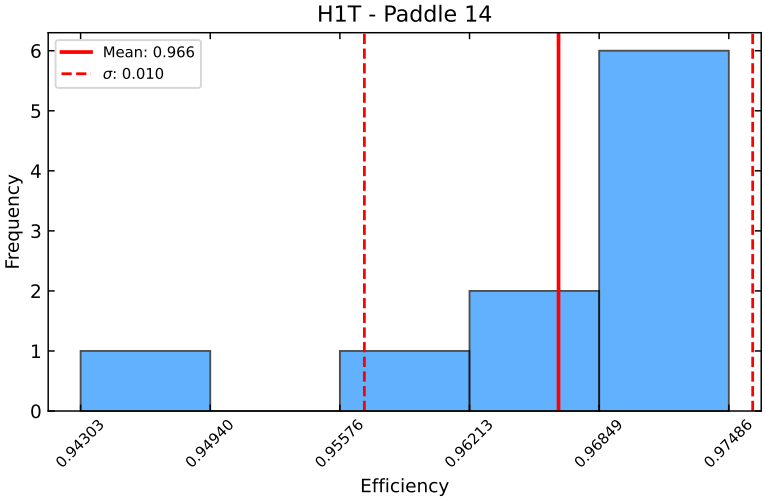
Efficiency Distribution: H1T - Paddle 12



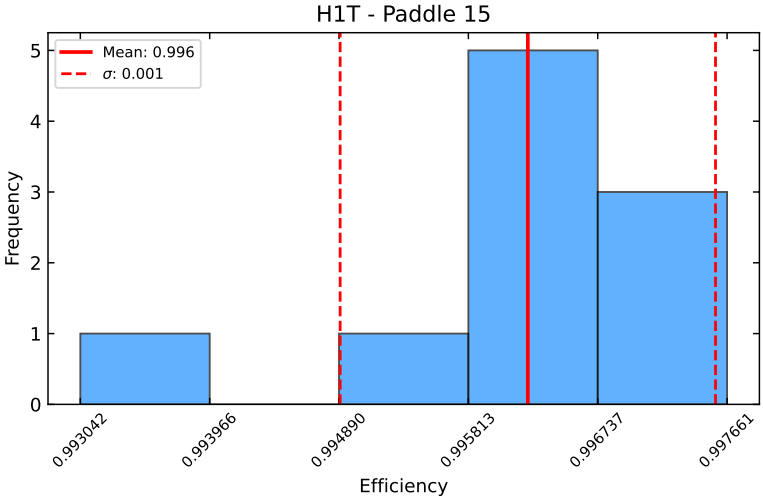
Efficiency Distribution: H1T - Paddle 13



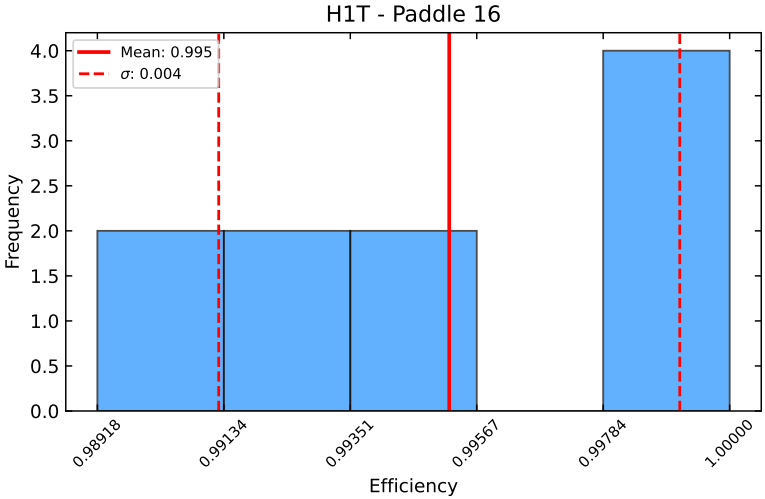
Efficiency Distribution: H1T - Paddle 14



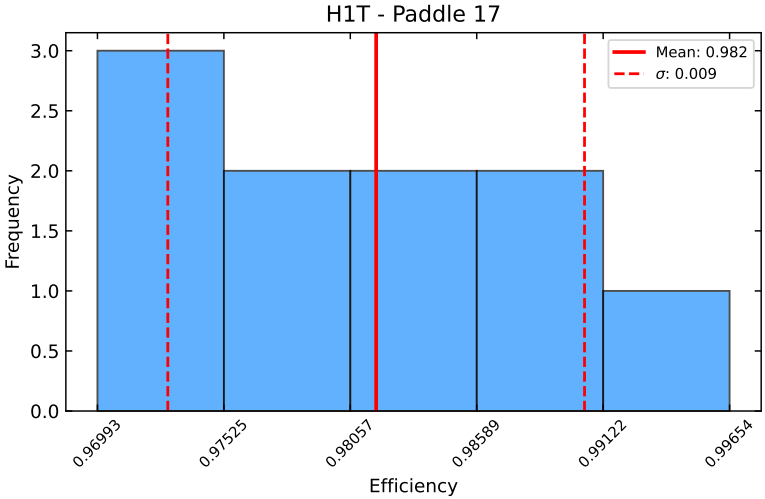
Efficiency Distribution: H1T - Paddle 15



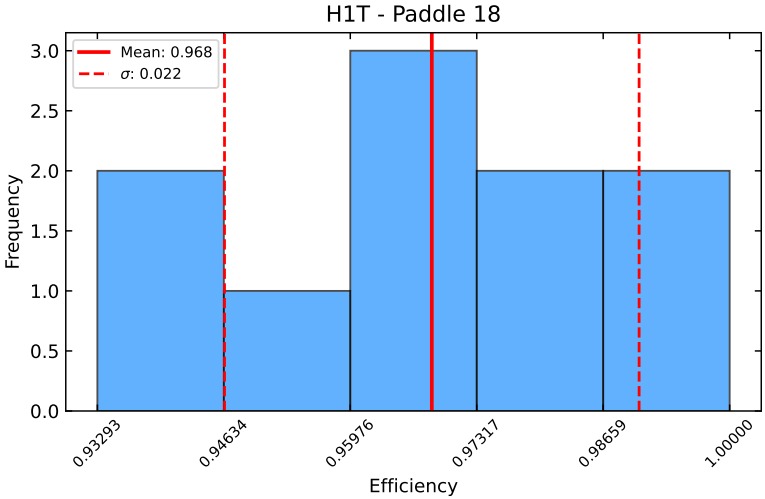
Efficiency Distribution: H1T - Paddle 16



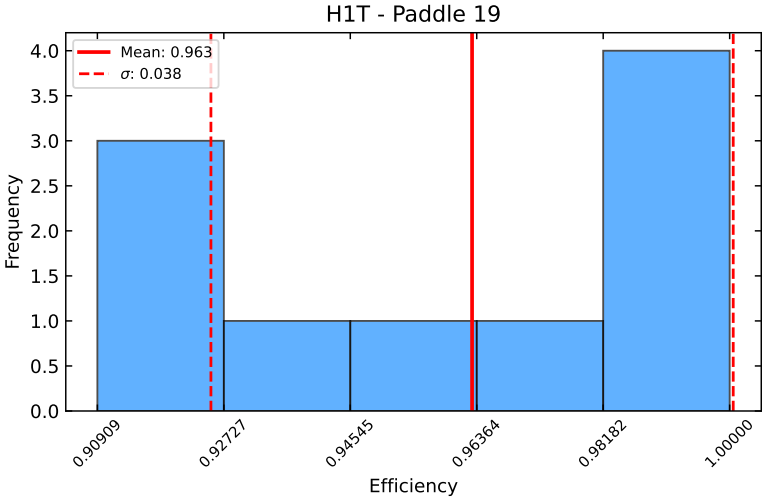
Efficiency Distribution: H1T - Paddle 17



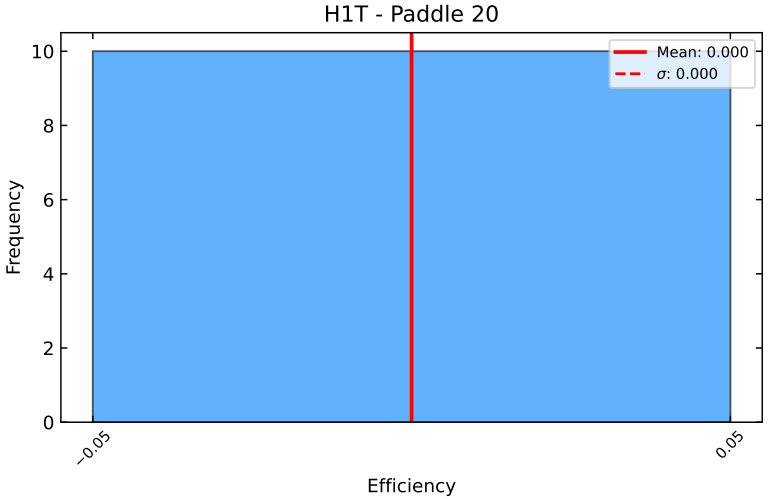
Efficiency Distribution: H1T - Paddle 18



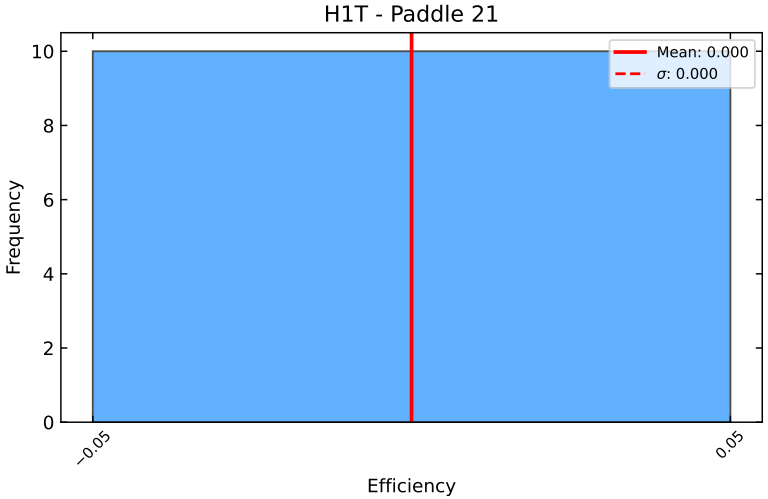
Efficiency Distribution: H1T - Paddle 19



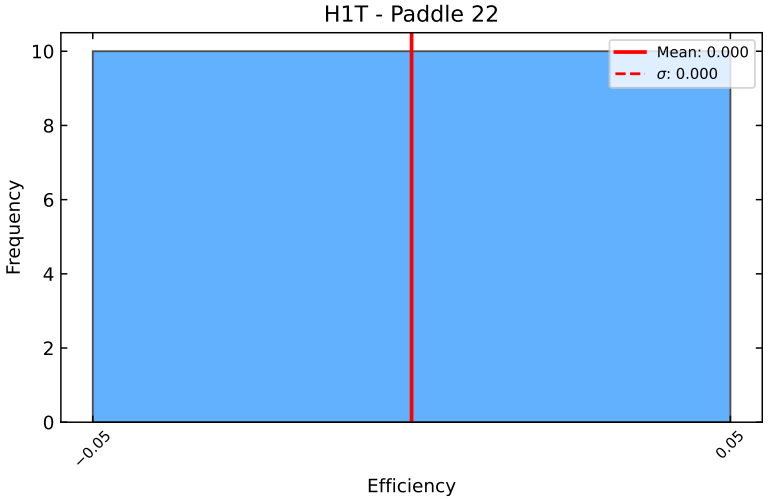
Efficiency Distribution: H1T - Paddle 20



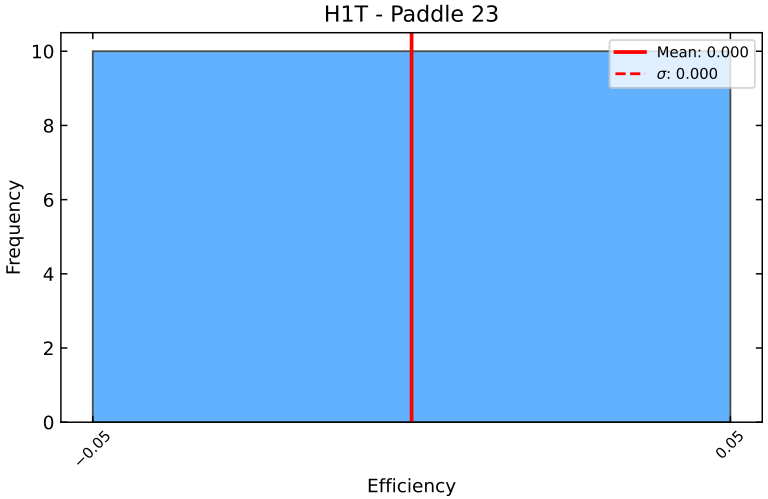
Efficiency Distribution: H1T - Paddle 21



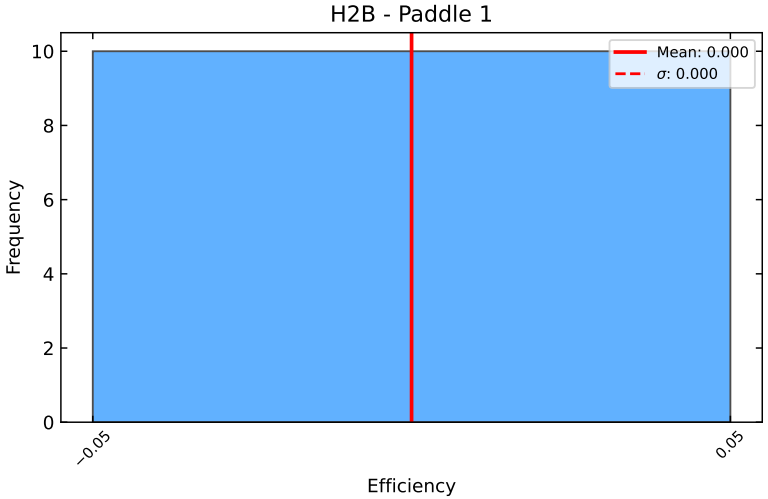
Efficiency Distribution: H1T - Paddle 22



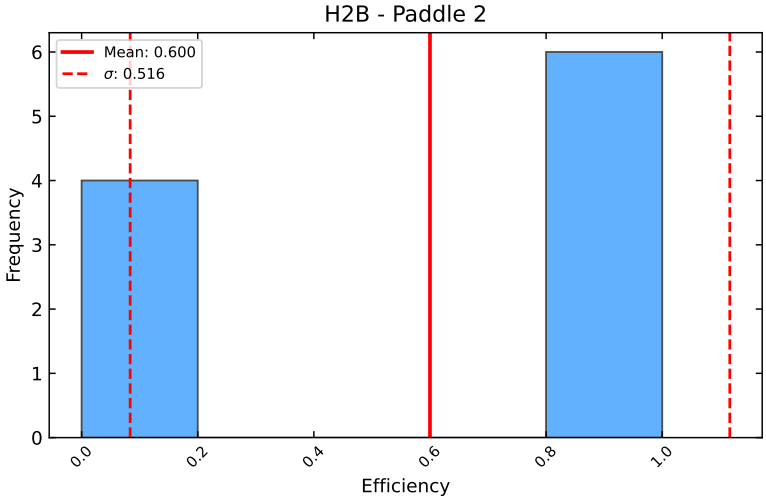
Efficiency Distribution: H1T - Paddle 23



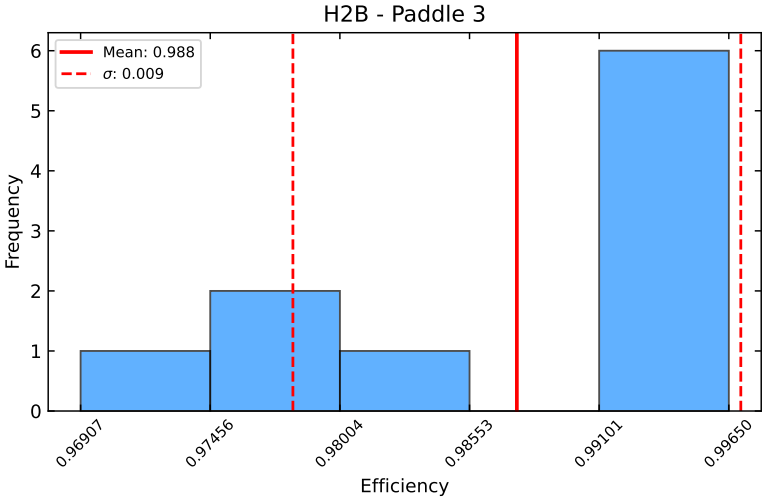
Efficiency Distribution: H2B - Paddle 1



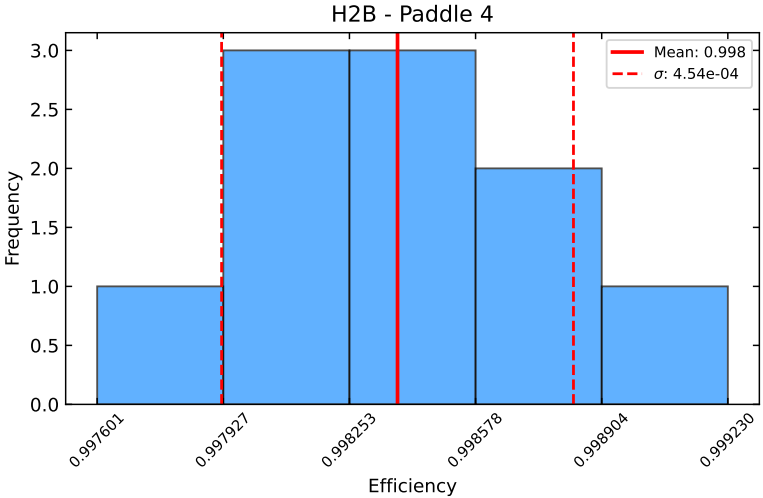
Efficiency Distribution: H2B - Paddle 2



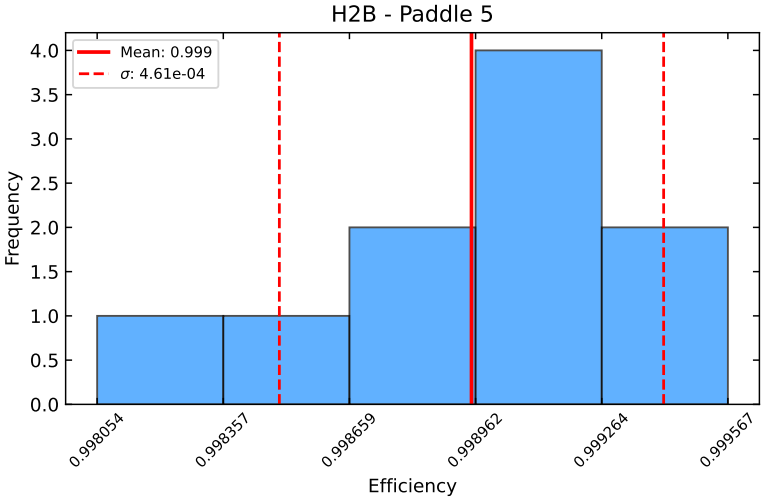
Efficiency Distribution: H2B - Paddle 3



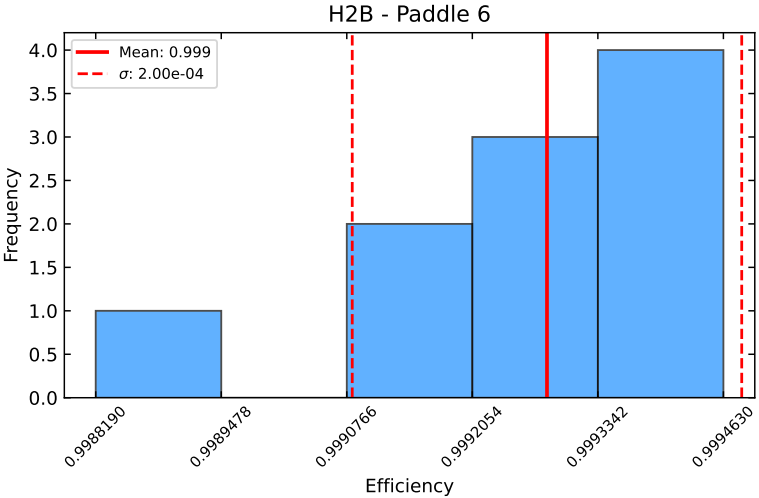
Efficiency Distribution: H2B - Paddle 4



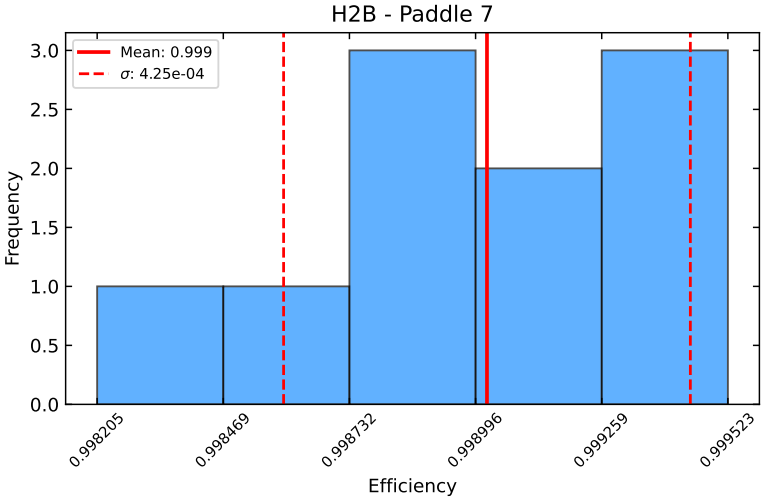
Efficiency Distribution: H2B - Paddle 5



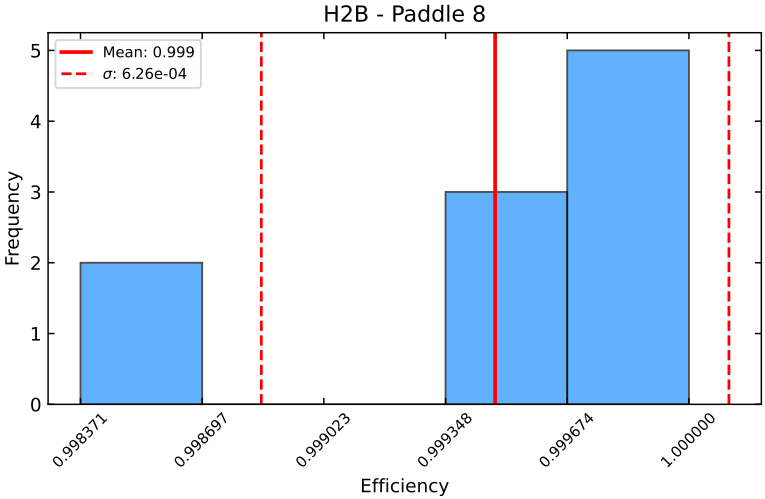
Efficiency Distribution: H2B - Paddle 6



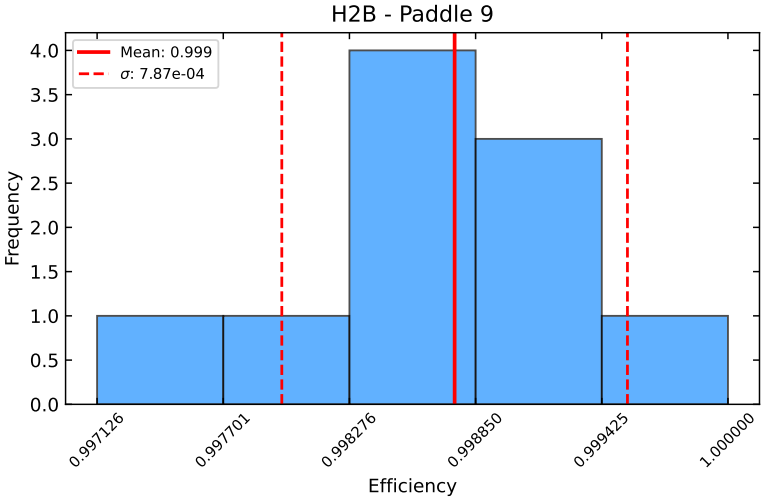
Efficiency Distribution: H2B - Paddle 7



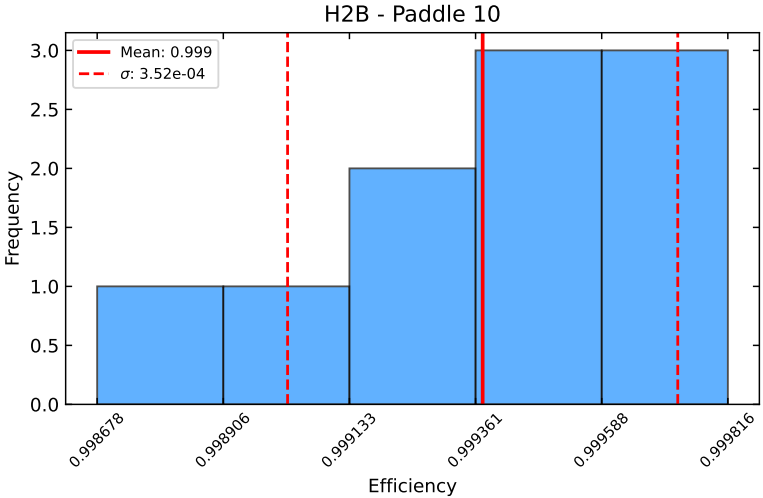
Efficiency Distribution: H2B - Paddle 8



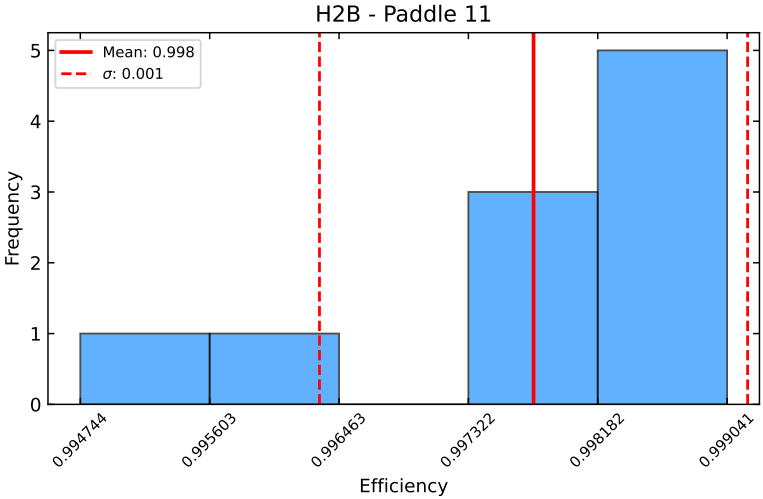
Efficiency Distribution: H2B - Paddle 9



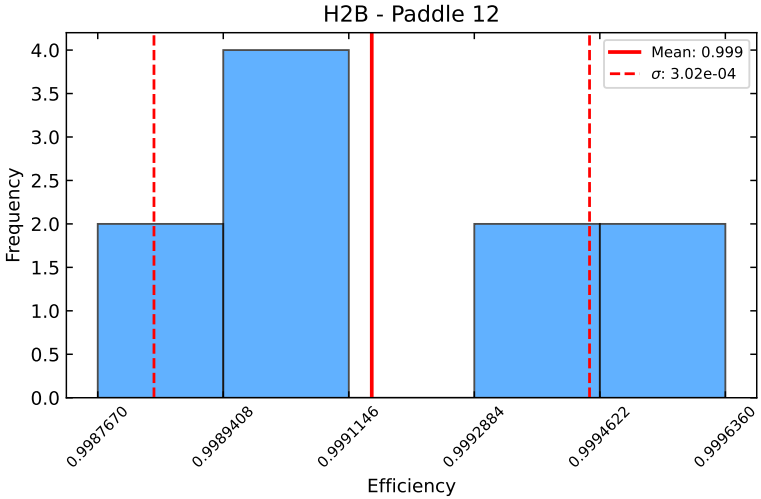
Efficiency Distribution: H2B - Paddle 10



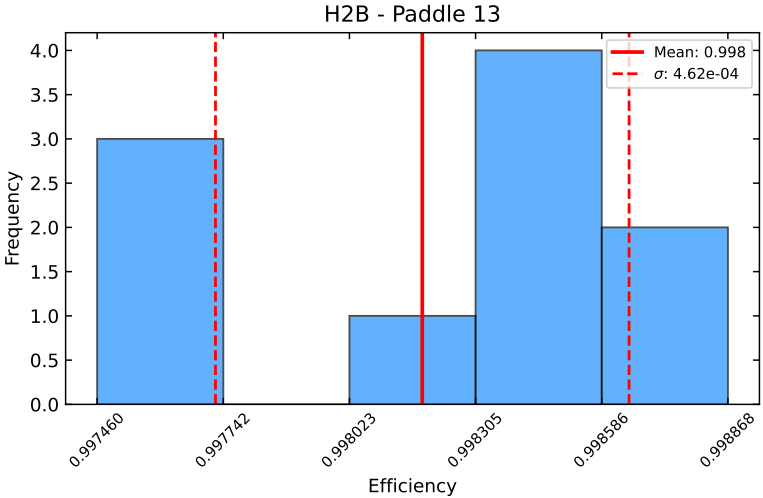
Efficiency Distribution: H2B - Paddle 11



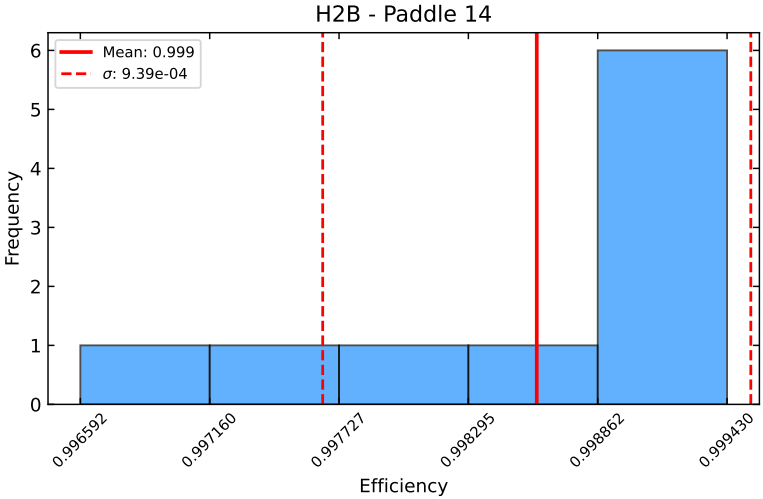
Efficiency Distribution: H2B - Paddle 12



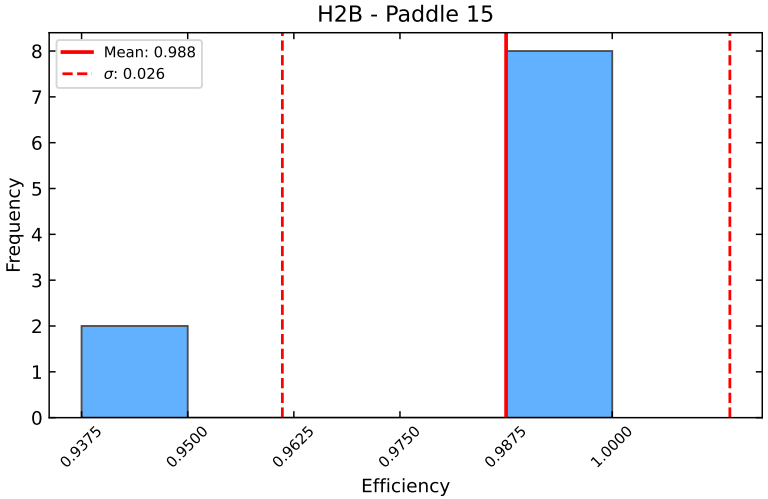
Efficiency Distribution: H2B - Paddle 13



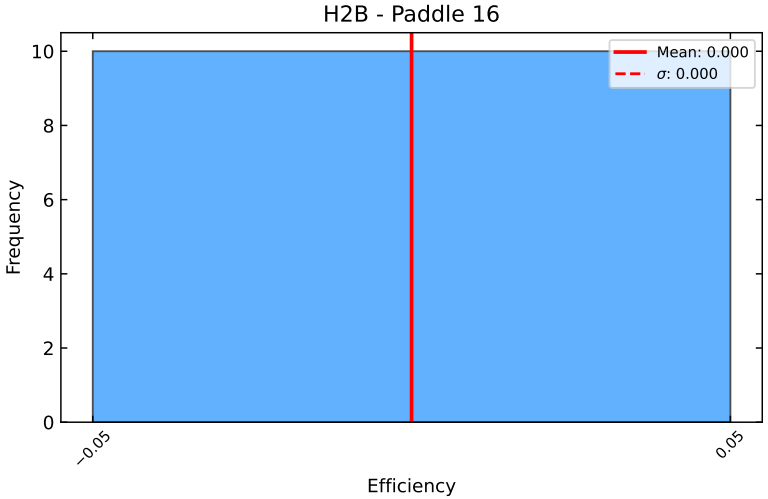
Efficiency Distribution: H2B - Paddle 14



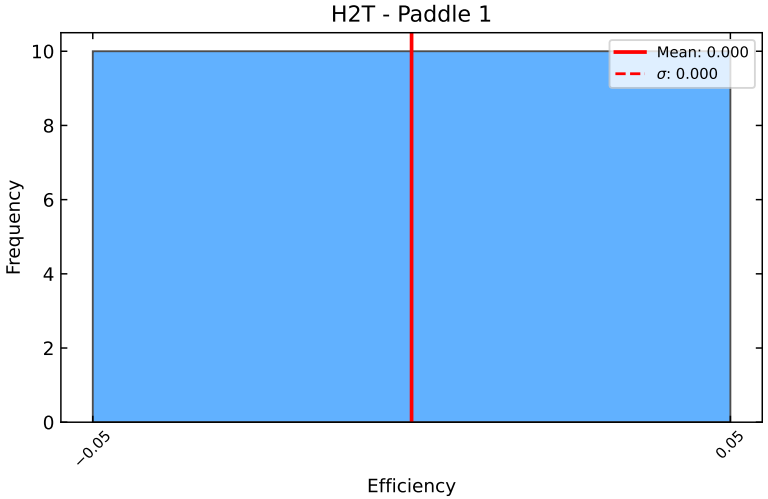
Efficiency Distribution: H2B - Paddle 15



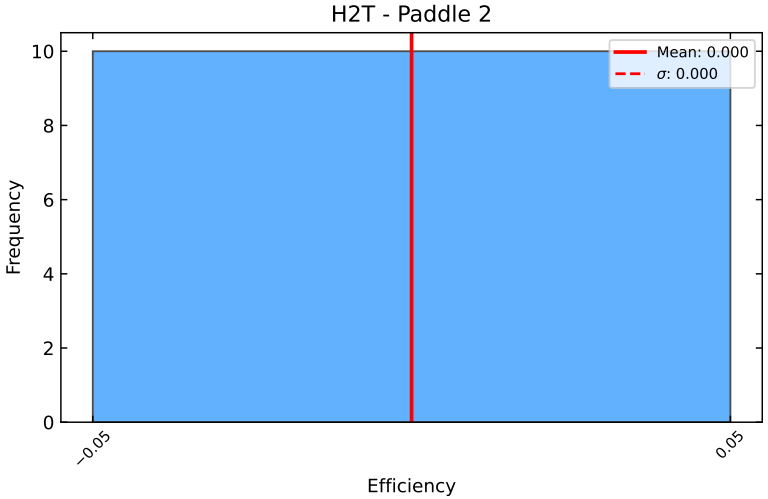
Efficiency Distribution: H2B - Paddle 16



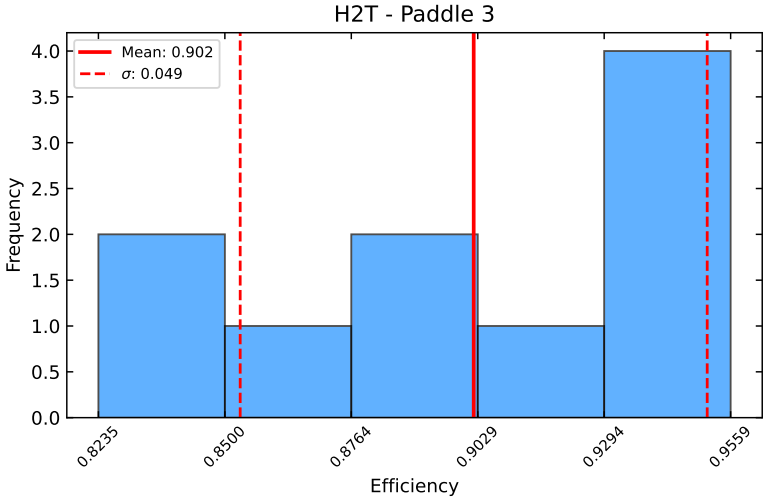
Efficiency Distribution: H2T - Paddle 1



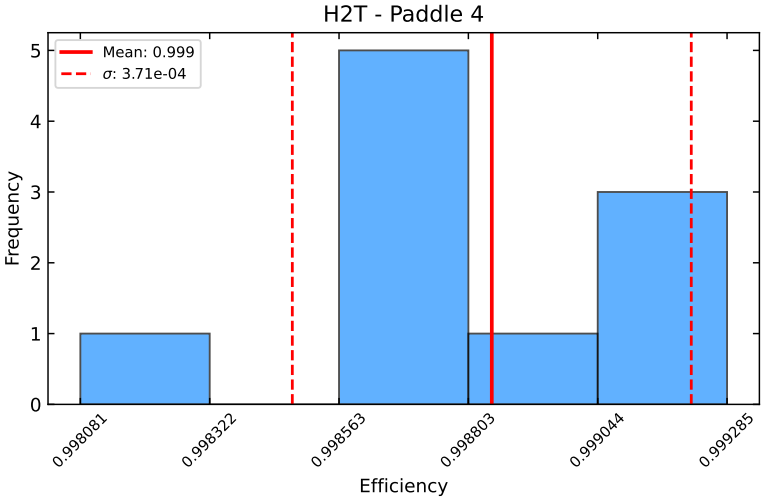
Efficiency Distribution: H2T - Paddle 2



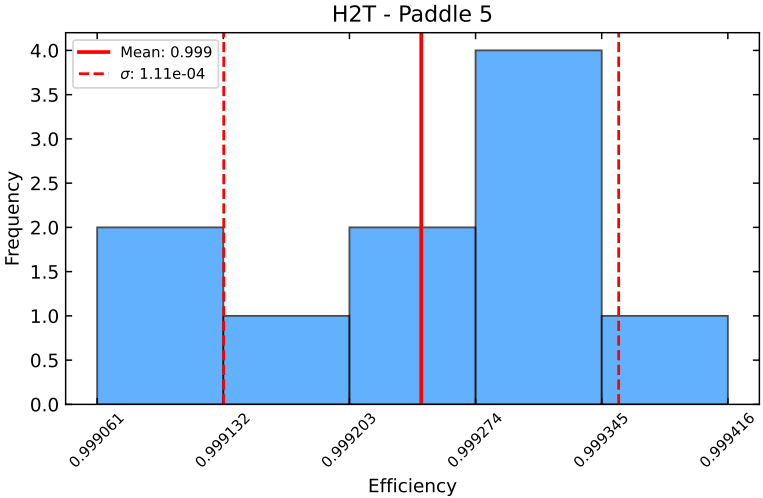
Efficiency Distribution: H2T - Paddle 3



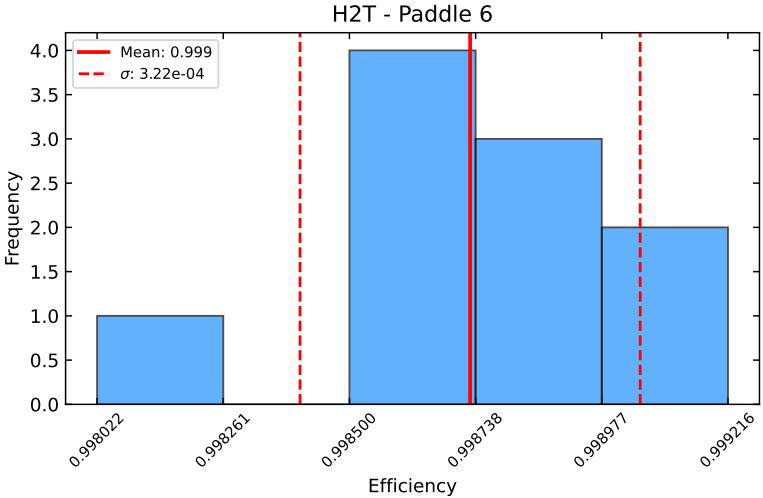
Efficiency Distribution: H2T - Paddle 4



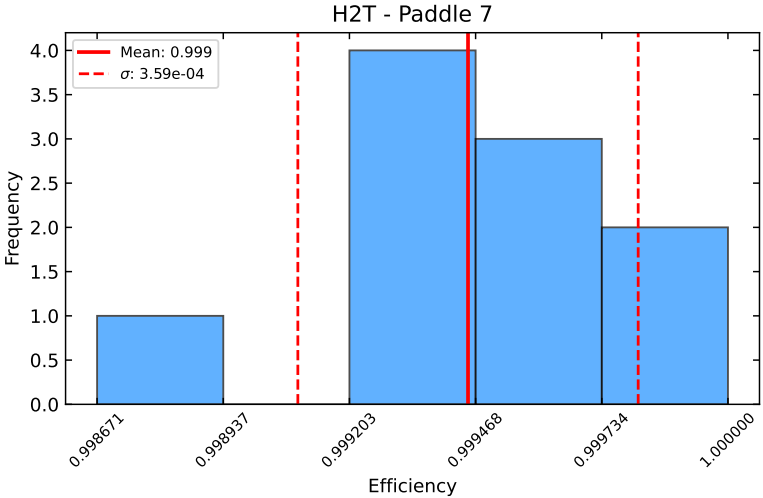
Efficiency Distribution: H2T - Paddle 5



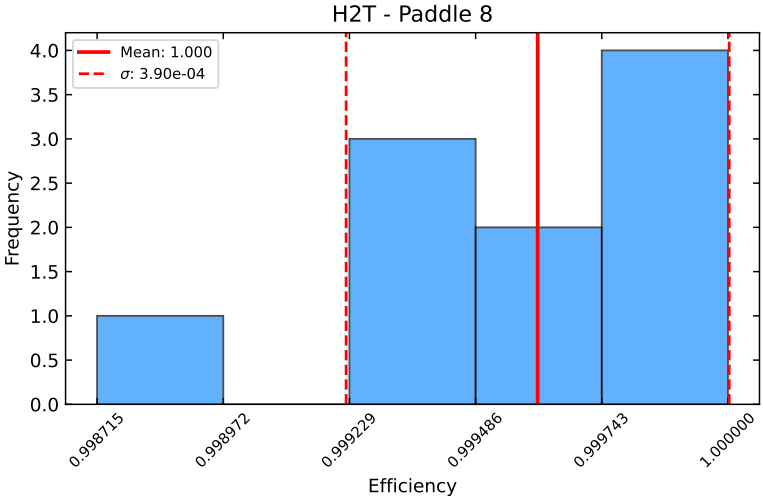
Efficiency Distribution: H2T - Paddle 6



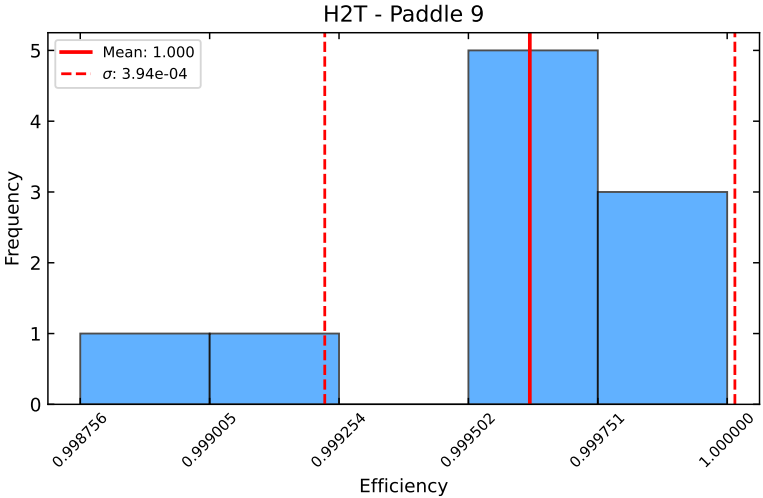
Efficiency Distribution: H2T - Paddle 7



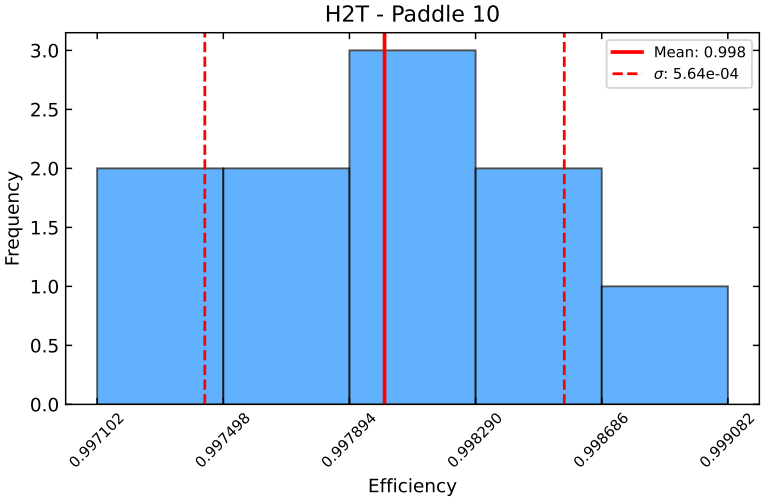
Efficiency Distribution: H2T - Paddle 8



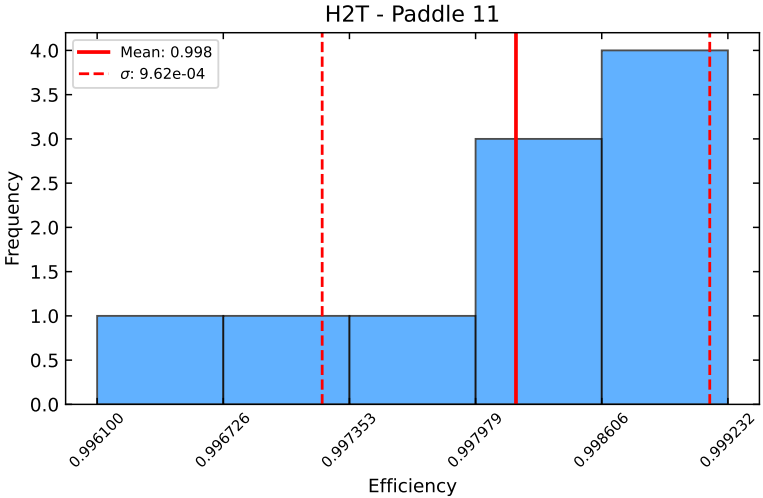
Efficiency Distribution: H2T - Paddle 9



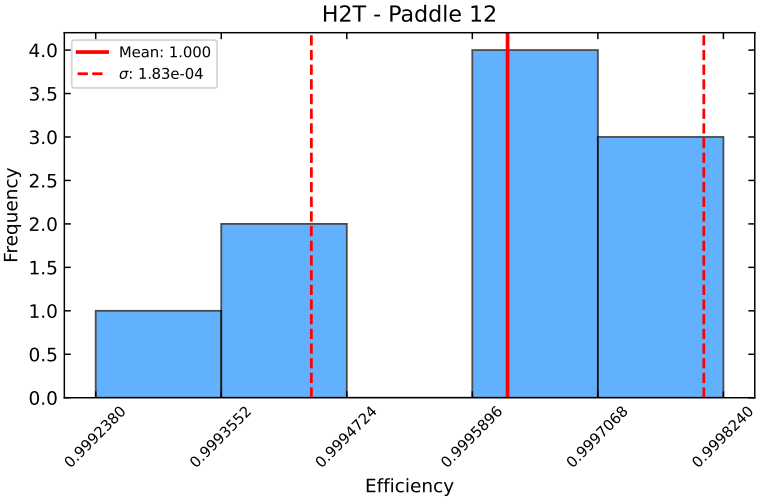
Efficiency Distribution: H2T - Paddle 10



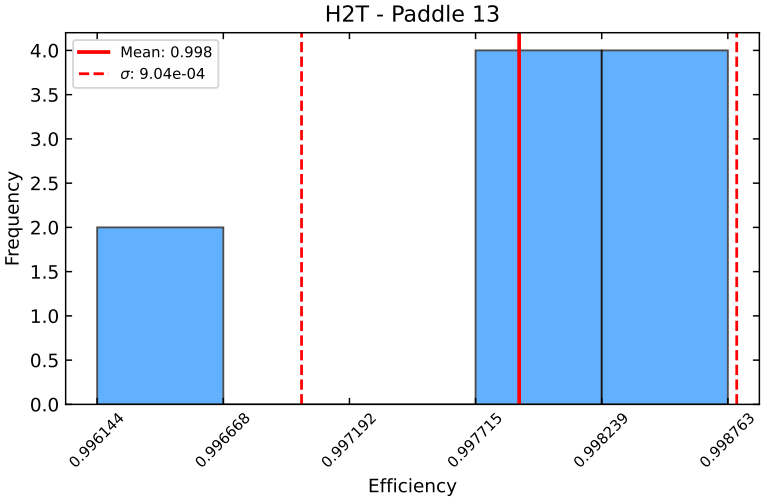
Efficiency Distribution: H2T - Paddle 11



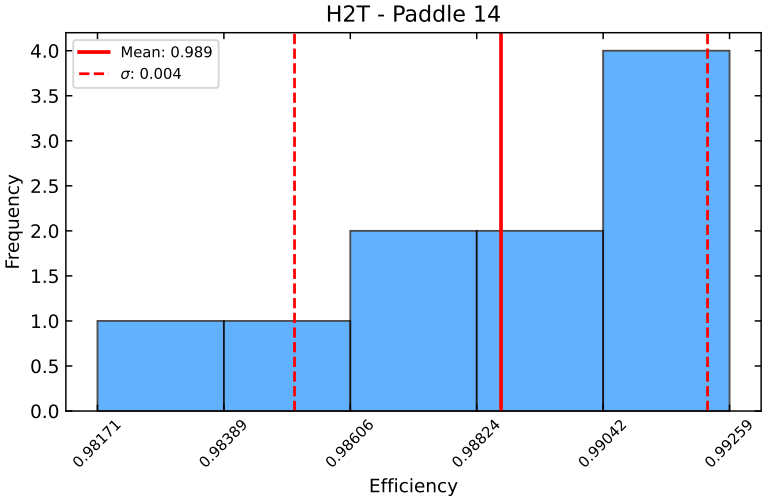
Efficiency Distribution: H2T - Paddle 12



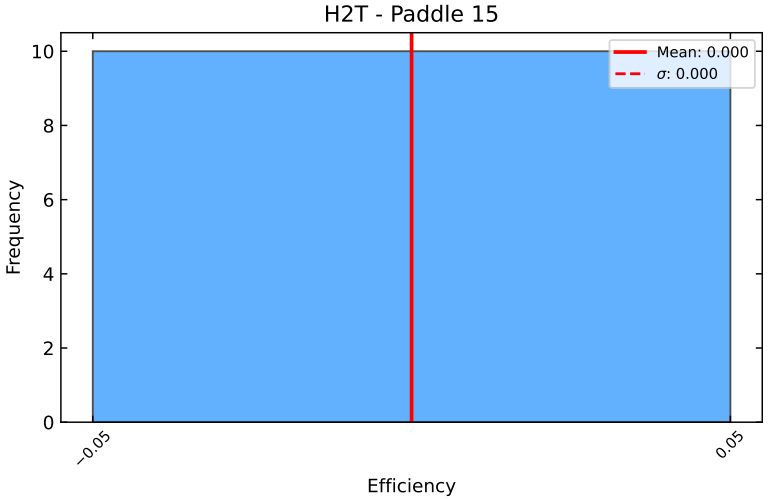
Efficiency Distribution: H2T - Paddle 13



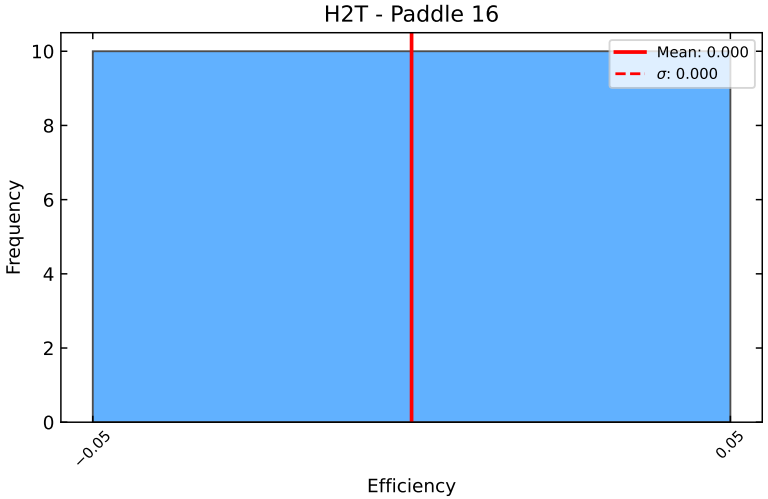
Efficiency Distribution: H2T - Paddle 14



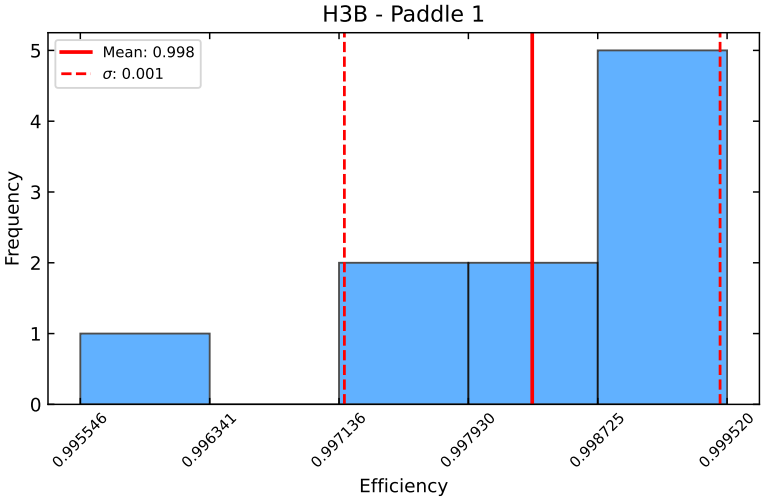
Efficiency Distribution: H2T - Paddle 15



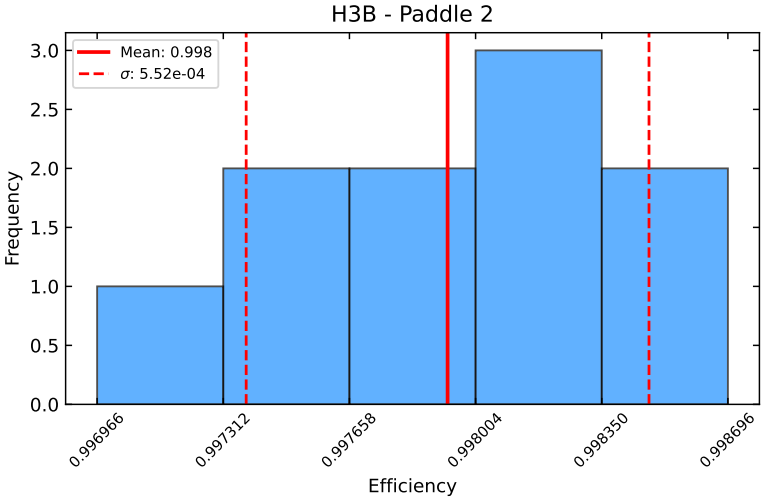
Efficiency Distribution: H2T - Paddle 16



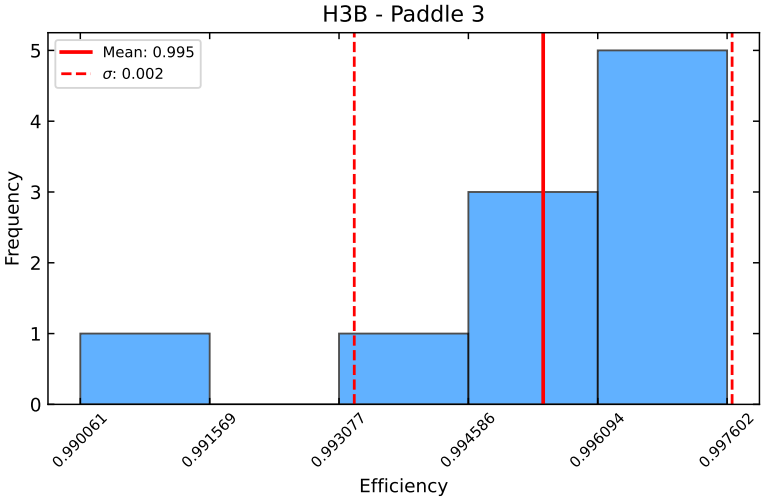
Efficiency Distribution: H3B - Paddle 1



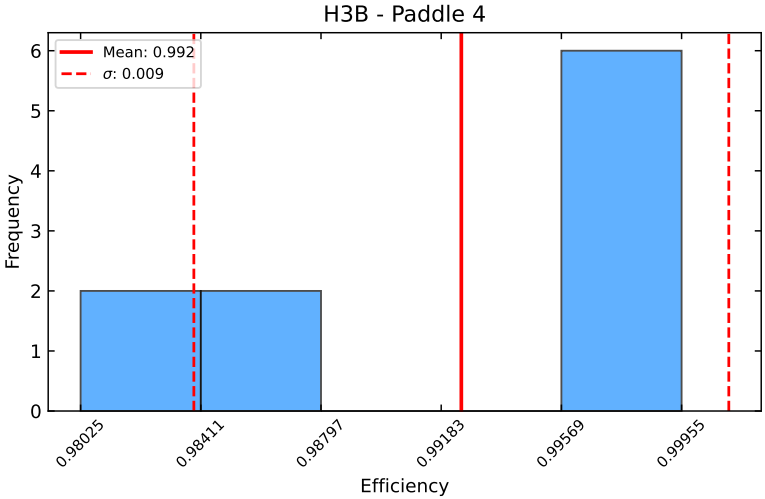
Efficiency Distribution: H3B - Paddle 2



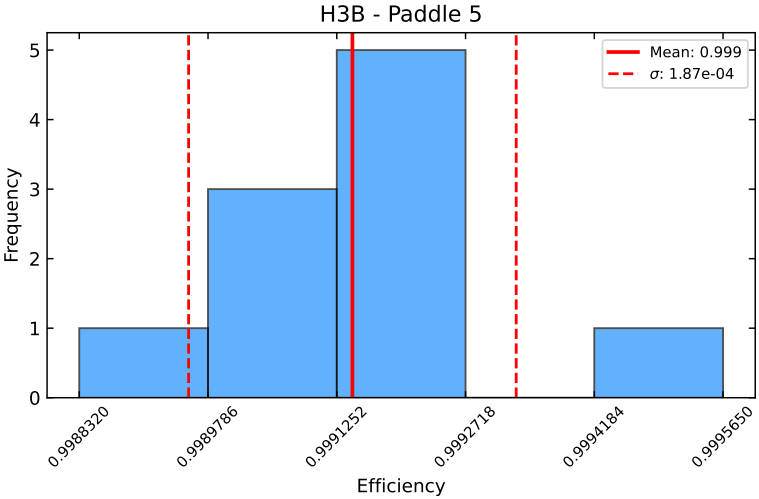
Efficiency Distribution: H3B - Paddle 3



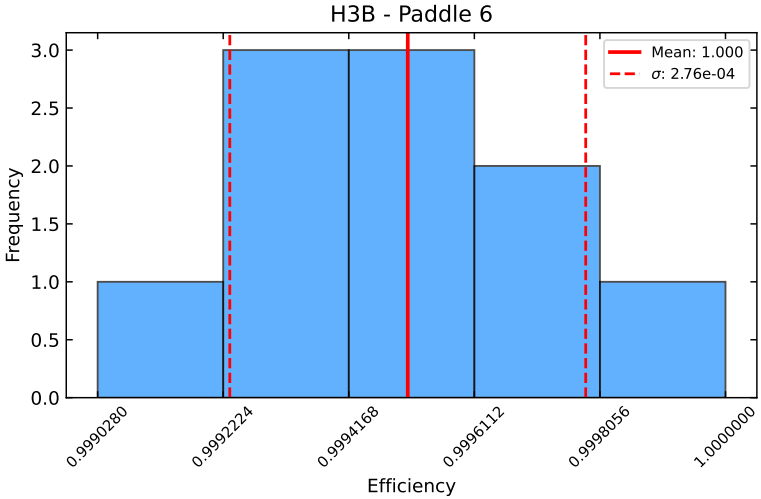
Efficiency Distribution: H3B - Paddle 4



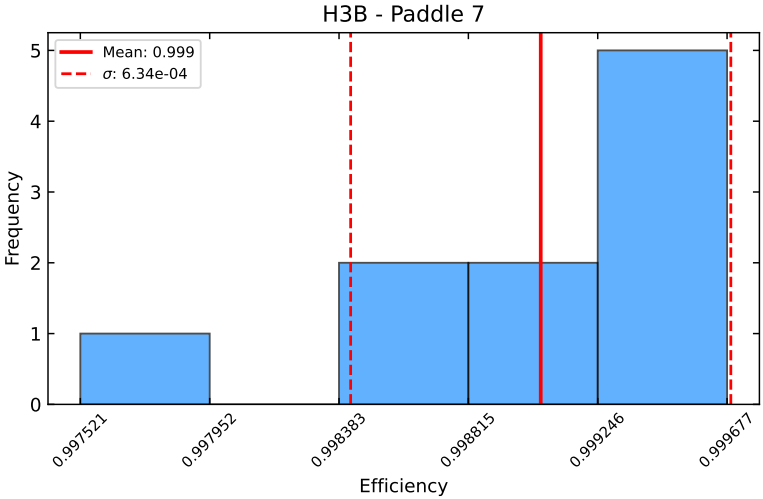
Efficiency Distribution: H3B - Paddle 5



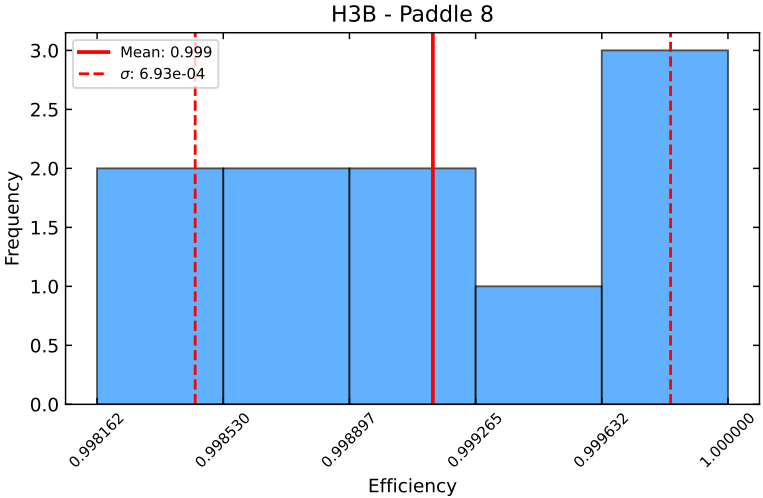
Efficiency Distribution: H3B - Paddle 6



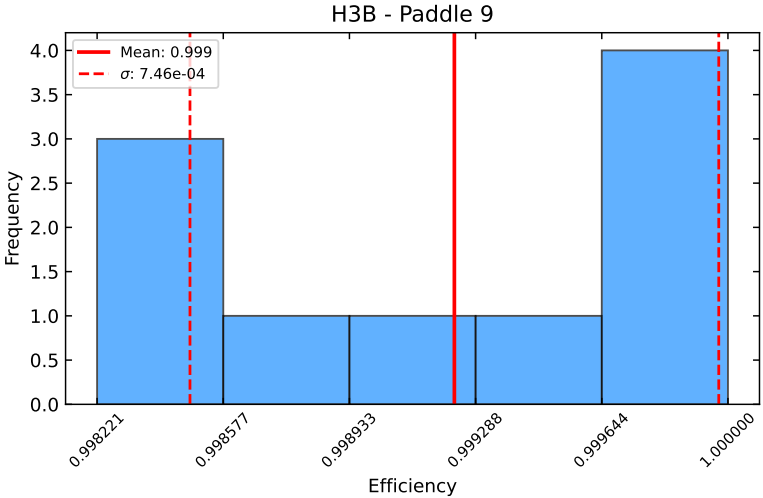
Efficiency Distribution: H3B - Paddle 7



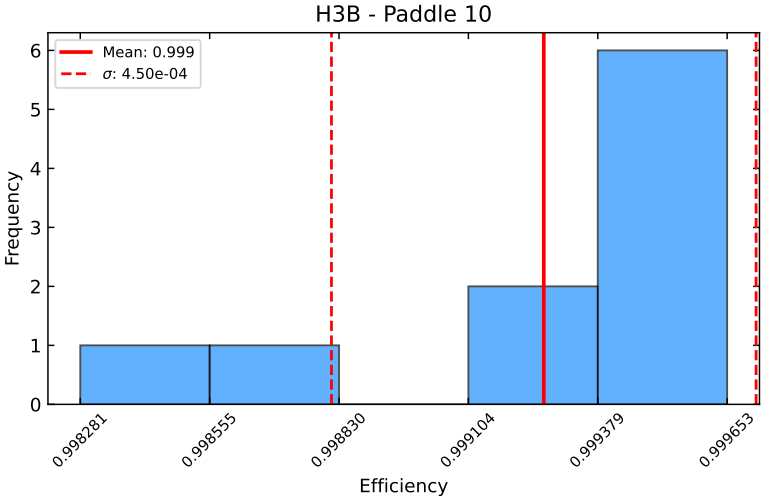
Efficiency Distribution: H3B - Paddle 8



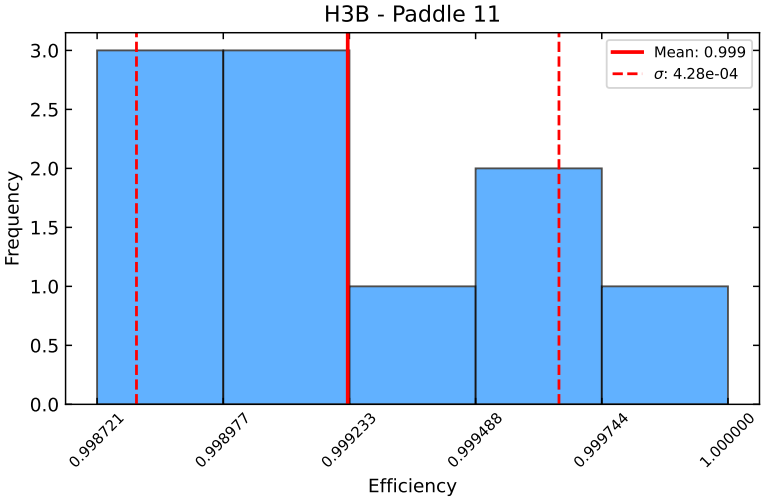
Efficiency Distribution: H3B - Paddle 9



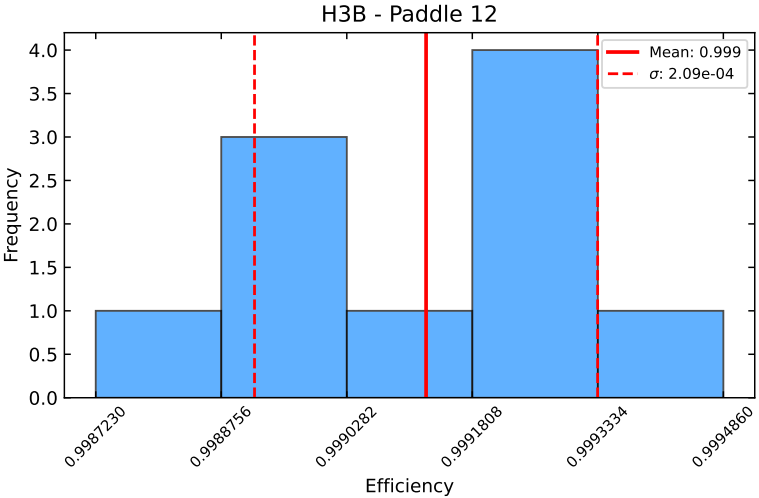
Efficiency Distribution: H3B - Paddle 10



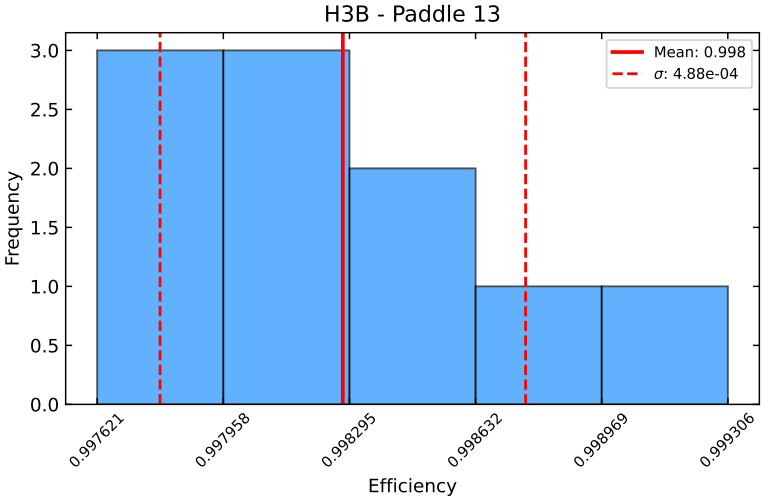
Efficiency Distribution: H3B - Paddle 11



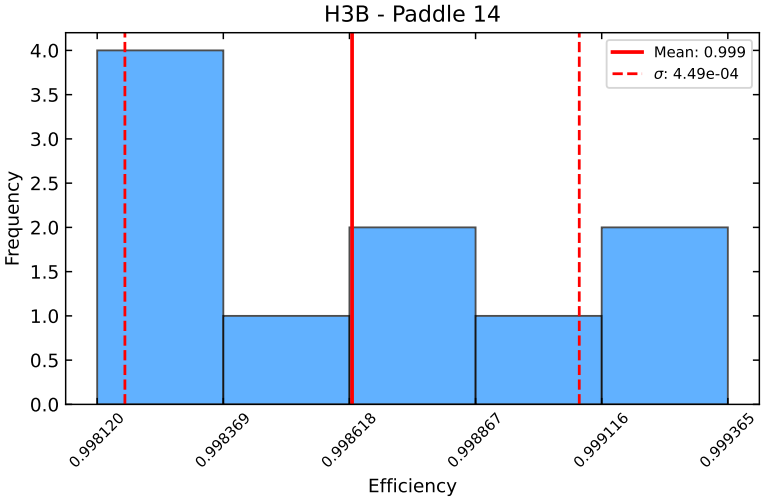
Efficiency Distribution: H3B - Paddle 12



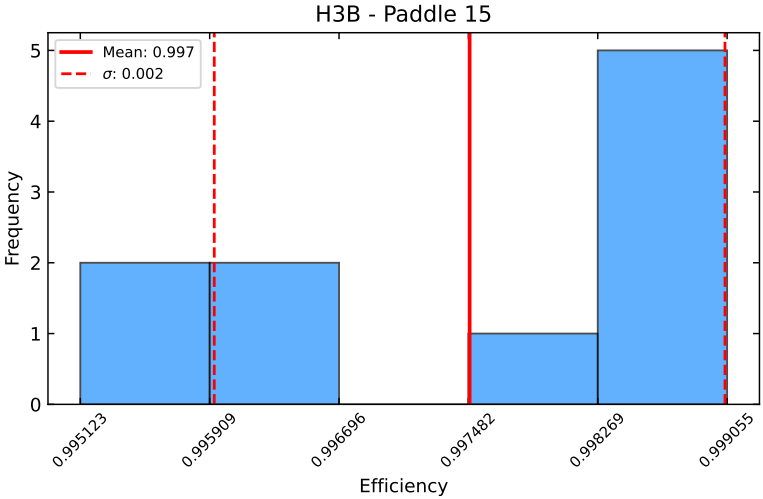
Efficiency Distribution: H3B - Paddle 13



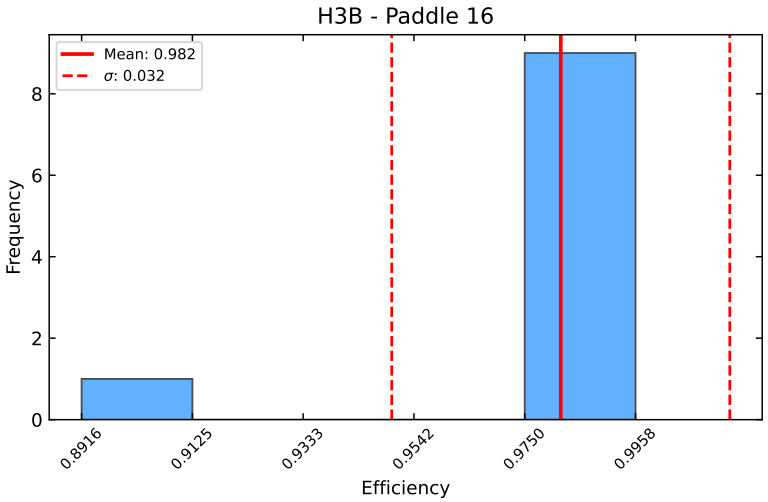
Efficiency Distribution: H3B - Paddle 14



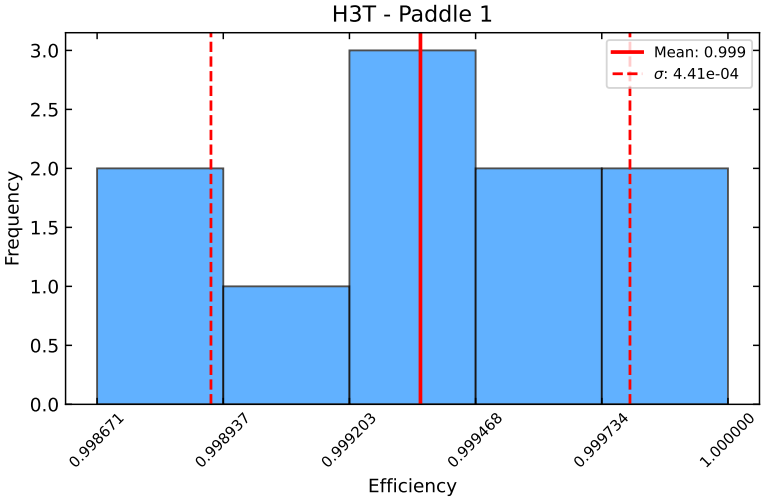
Efficiency Distribution: H3B - Paddle 15



Efficiency Distribution: H3B - Paddle 16



Efficiency Distribution: H3T - Paddle 1



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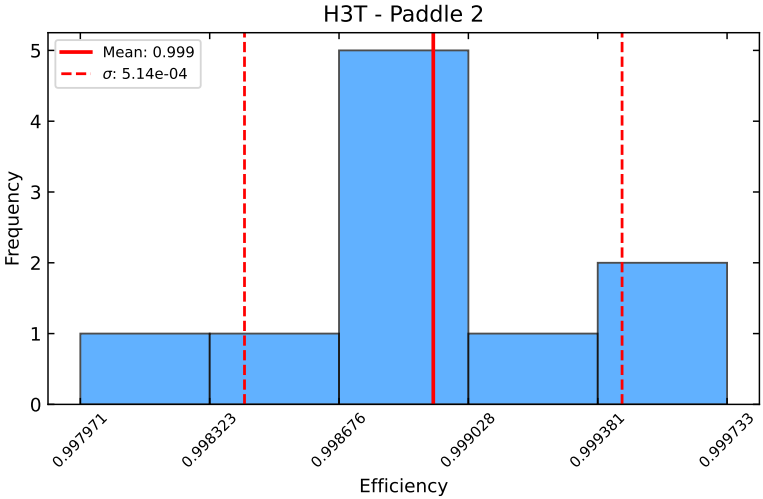

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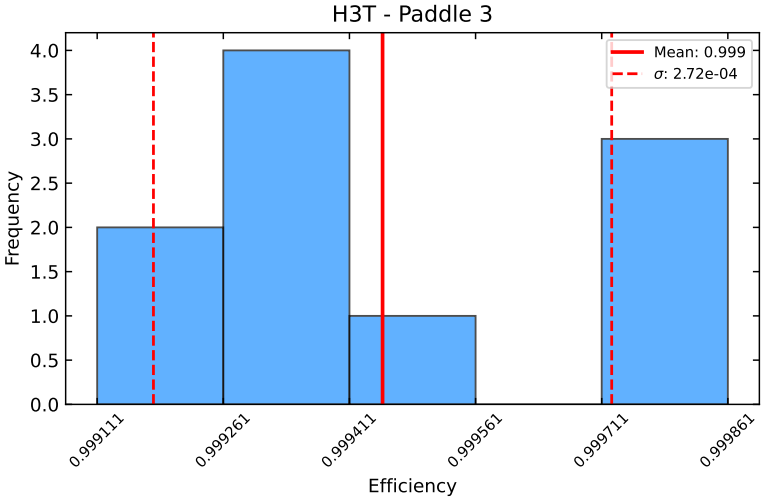

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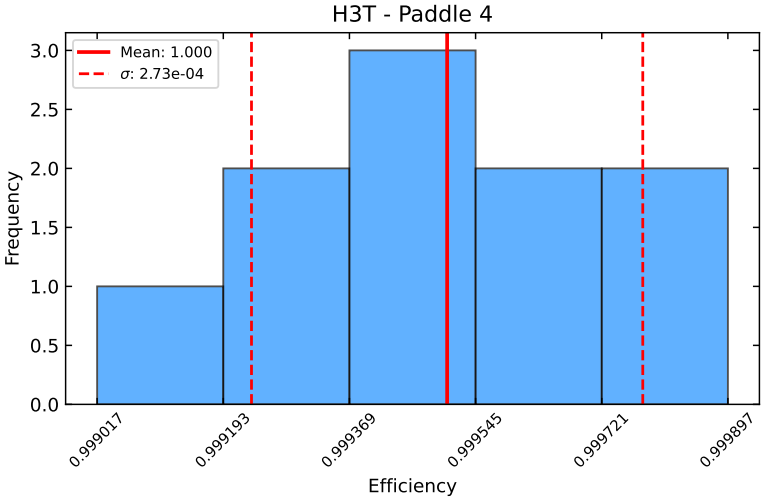

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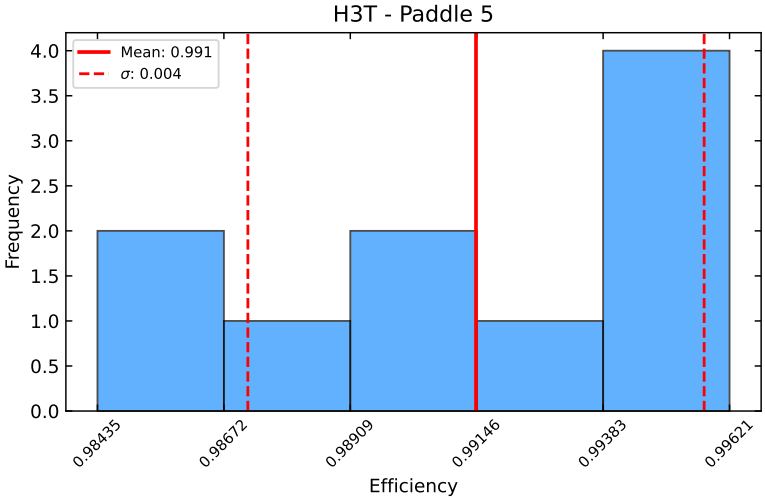
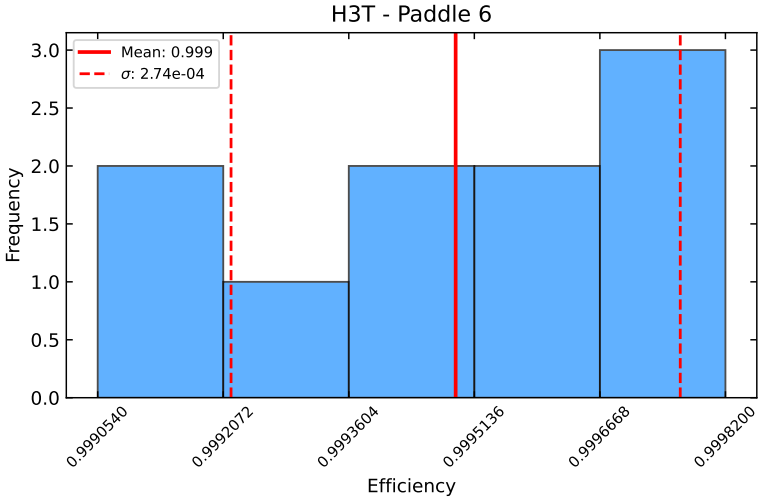
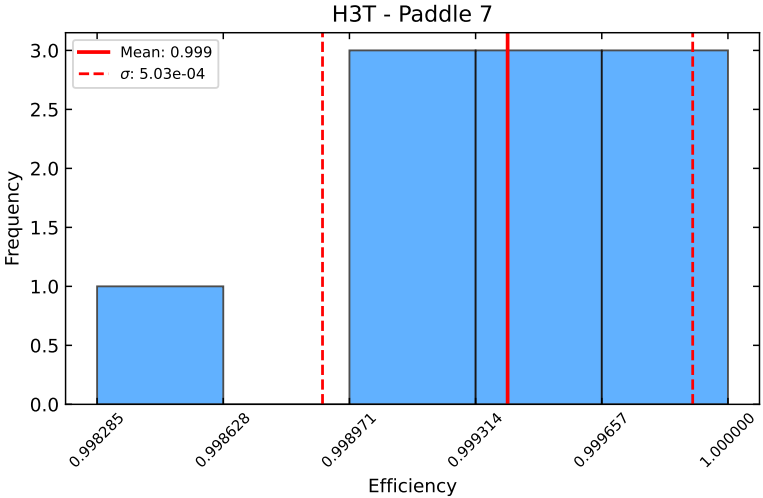


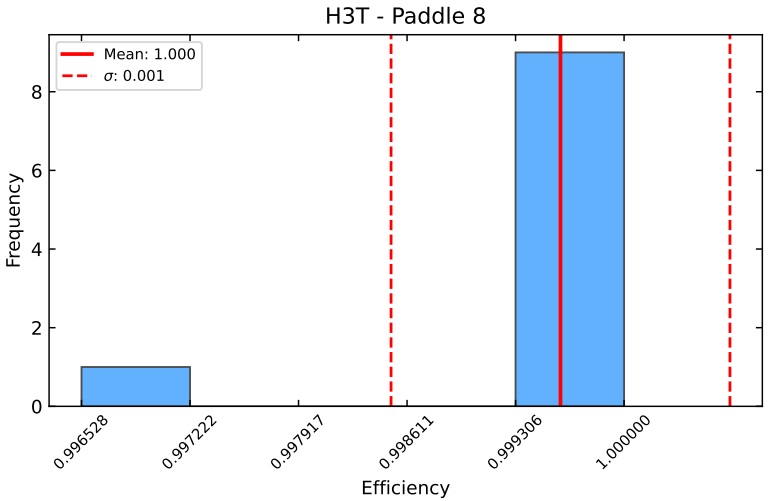
Efficiency Distribution: H3T - Paddle 6



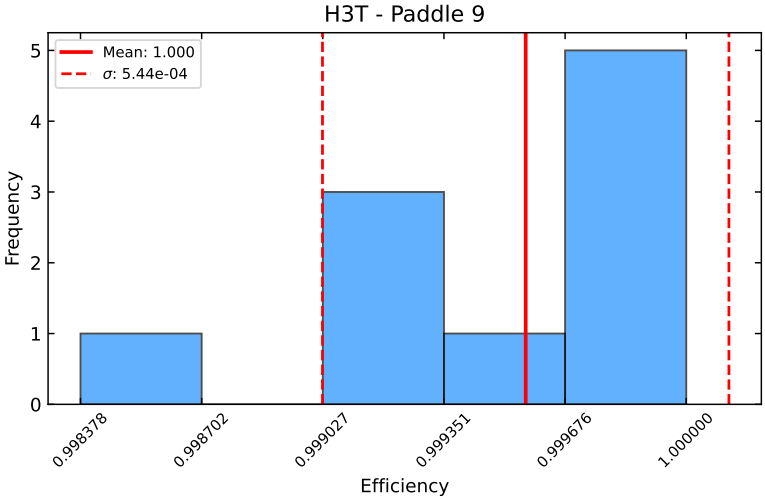
Efficiency Distribution: H3T - Paddle 7



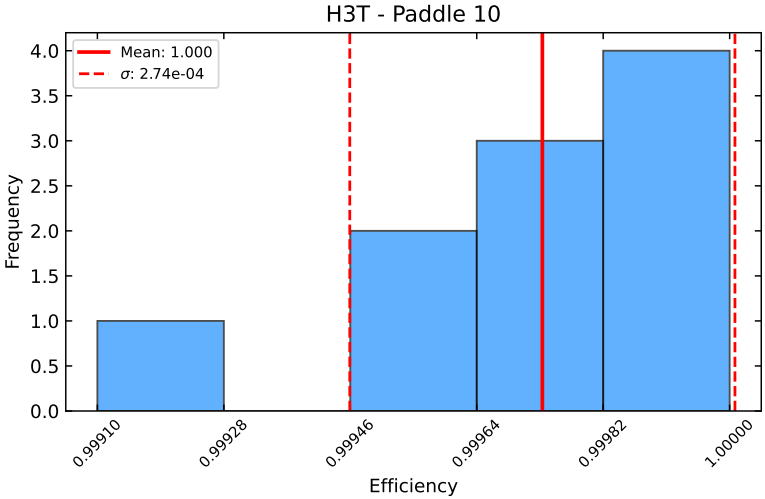
Efficiency Distribution: H3T - Paddle 8



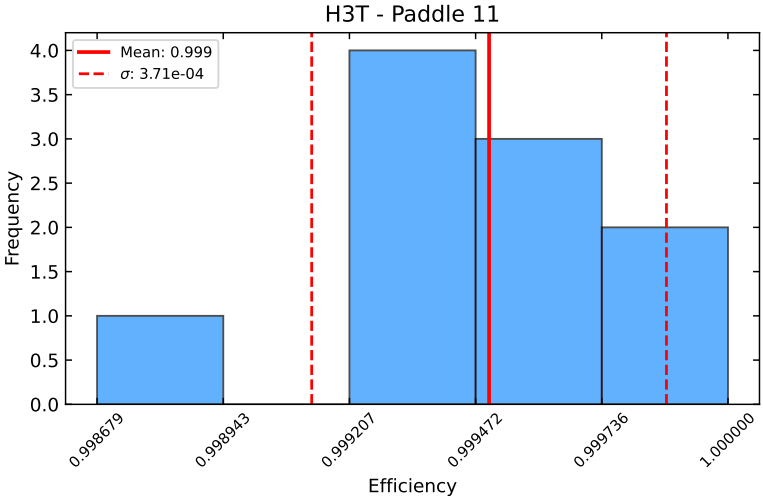
Efficiency Distribution: H3T - Paddle 9



Efficiency Distribution: H3T - Paddle 10



Efficiency Distribution: H3T - Paddle 11



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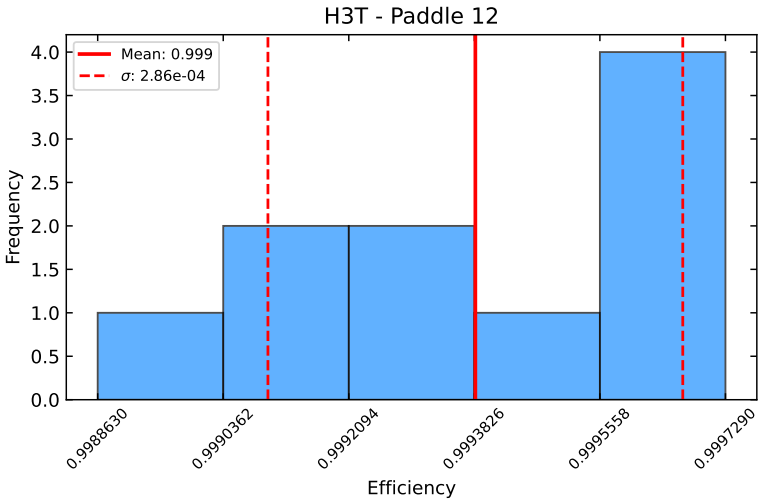
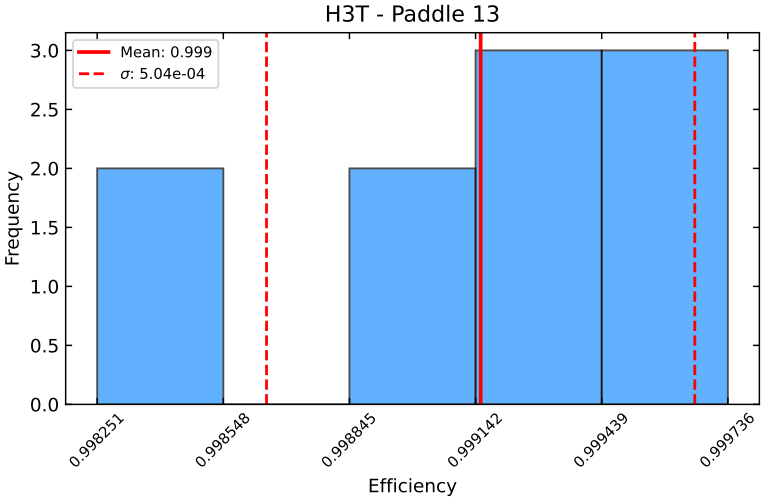
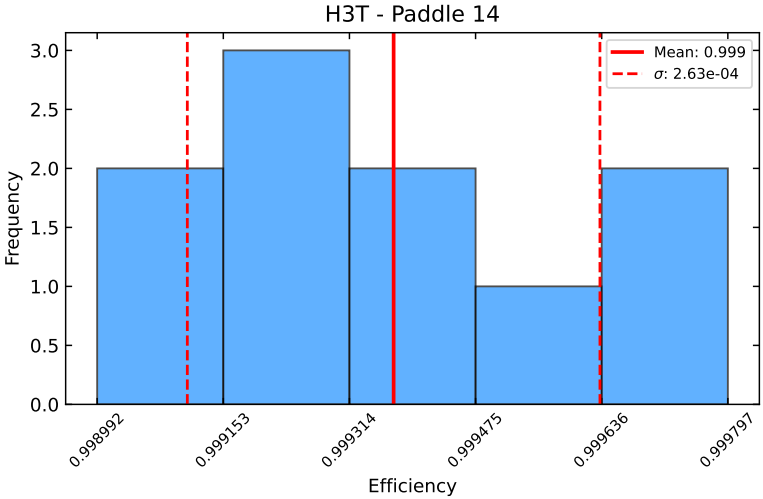


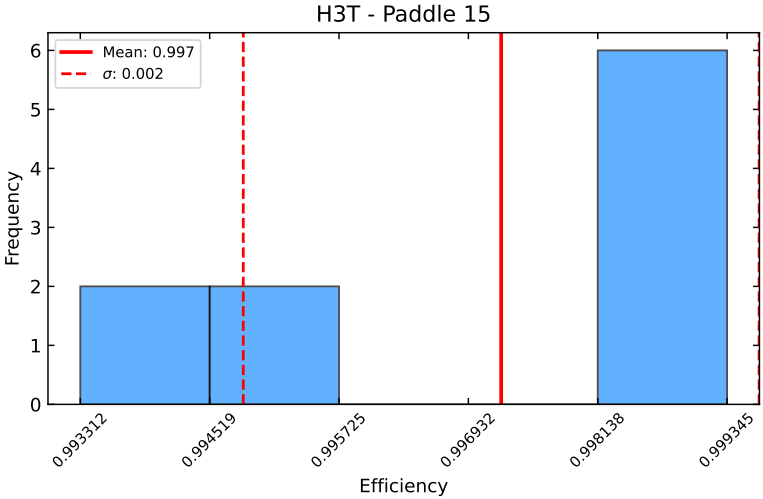
Efficiency Distribution: H3T - Paddle 13



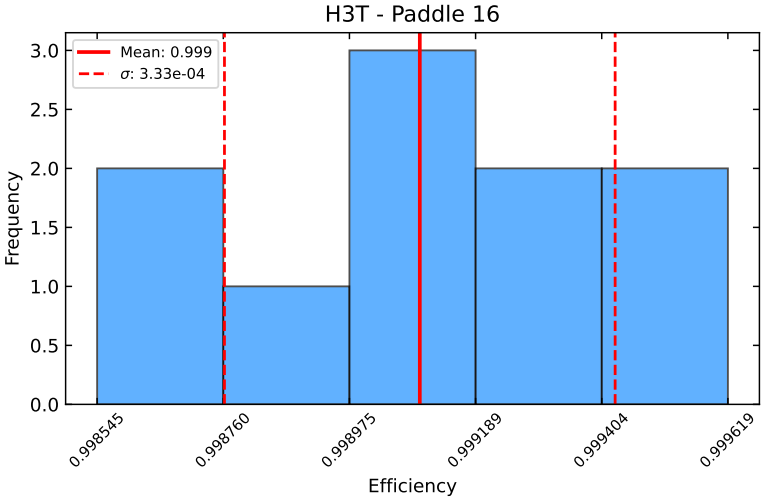
Efficiency Distribution: H3T - Paddle 14



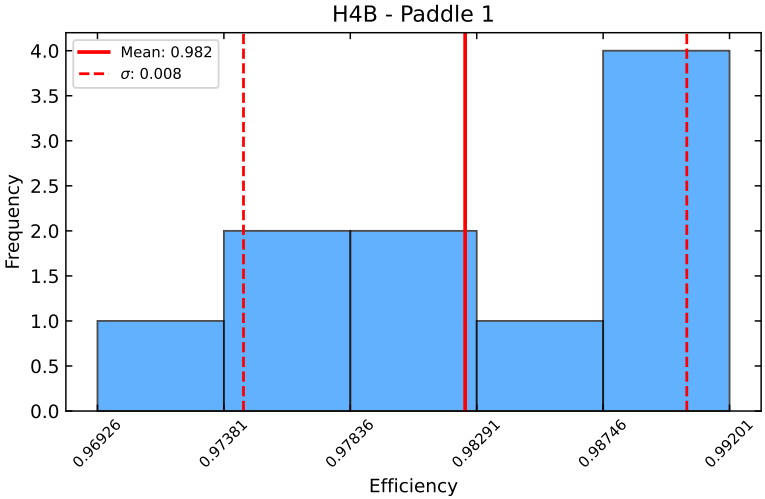
Efficiency Distribution: H3T - Paddle 15



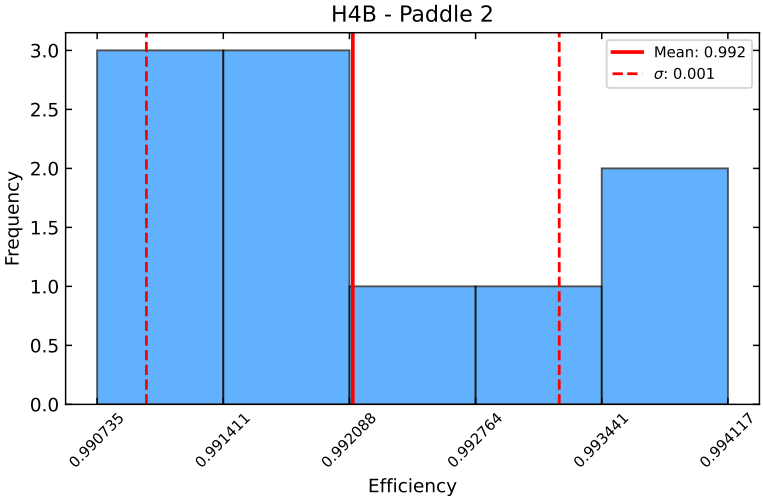
Efficiency Distribution: H3T - Paddle 16



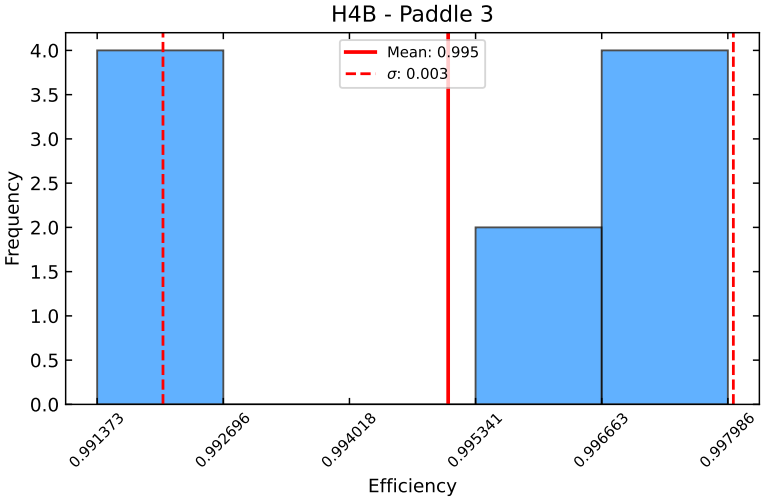
Efficiency Distribution: H4B - Paddle 1



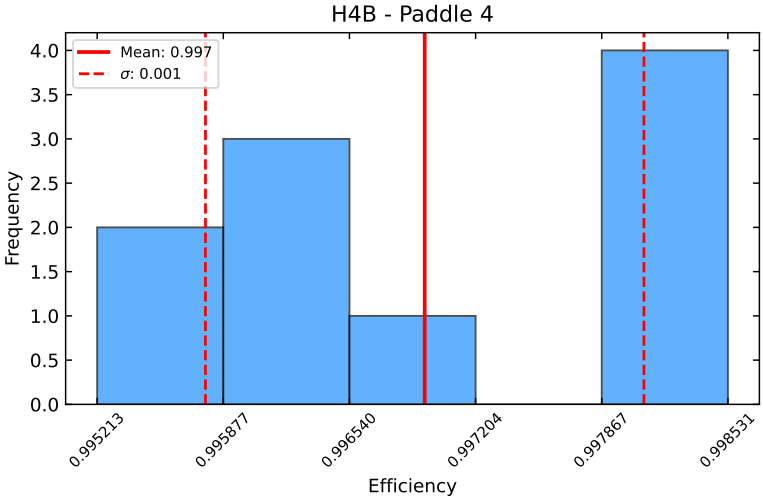
Efficiency Distribution: H4B - Paddle 2



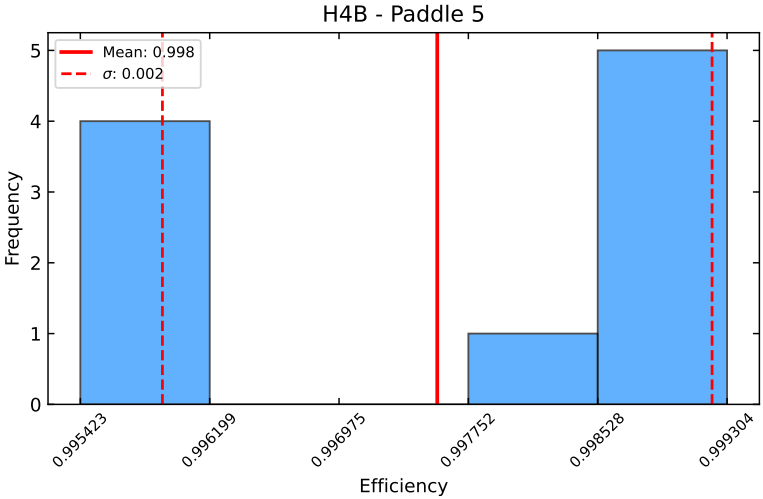
Efficiency Distribution: H4B - Paddle 3



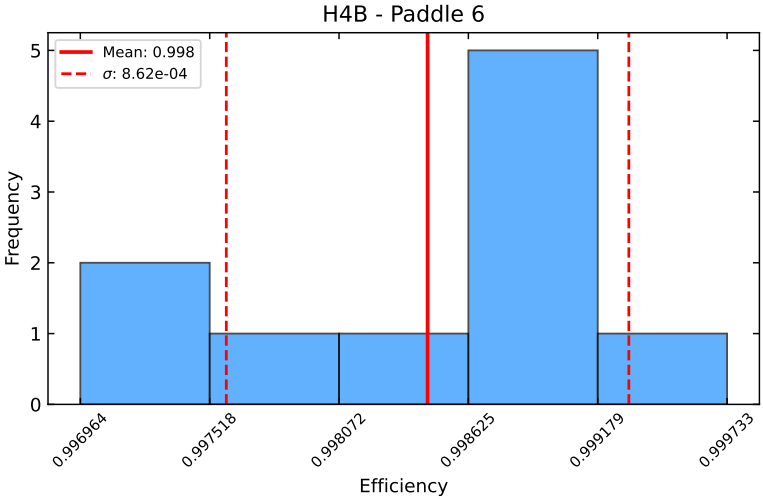
Efficiency Distribution: H4B - Paddle 4



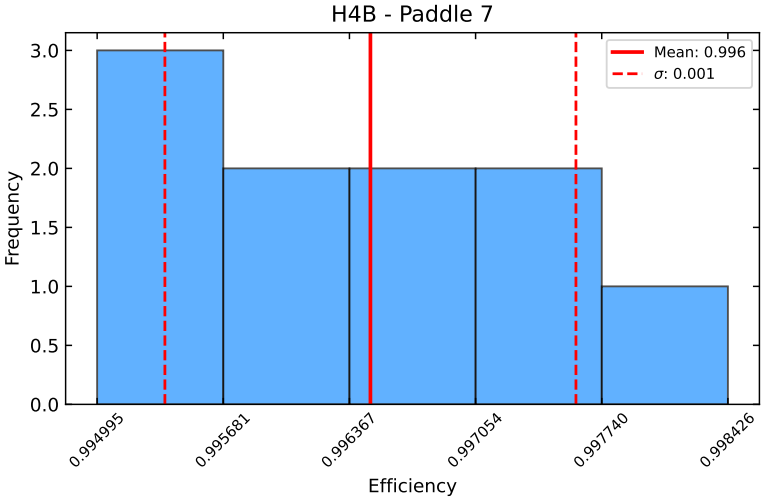
Efficiency Distribution: H4B - Paddle 5



Efficiency Distribution: H4B - Paddle 6



Efficiency Distribution: H4B - Paddle 7



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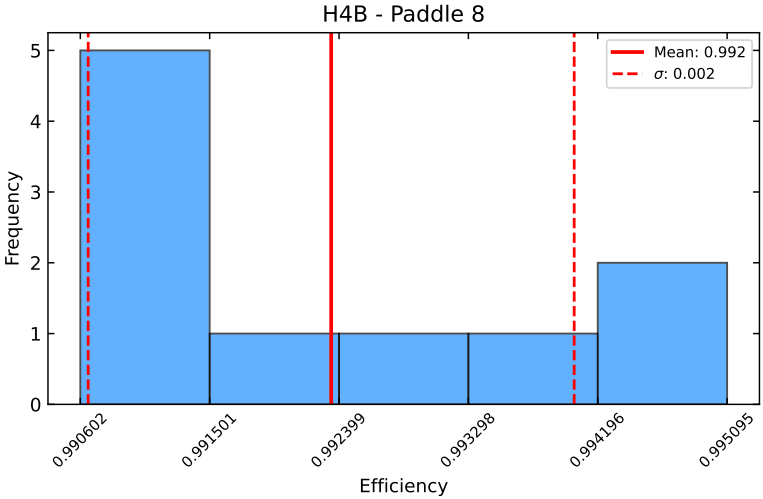
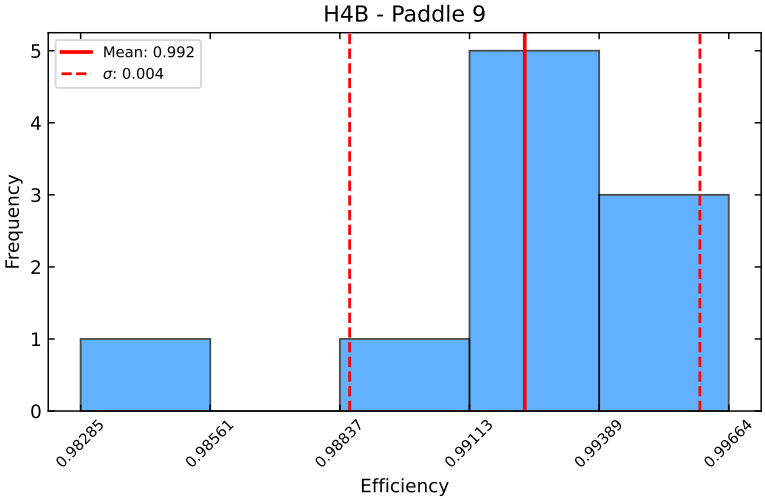
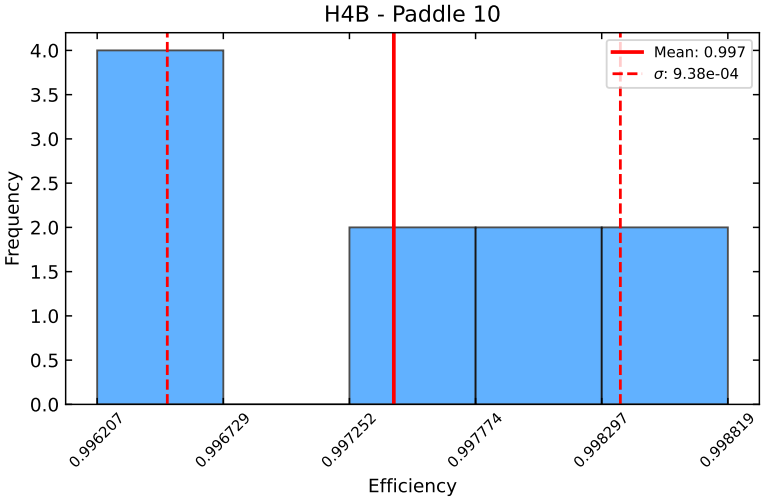


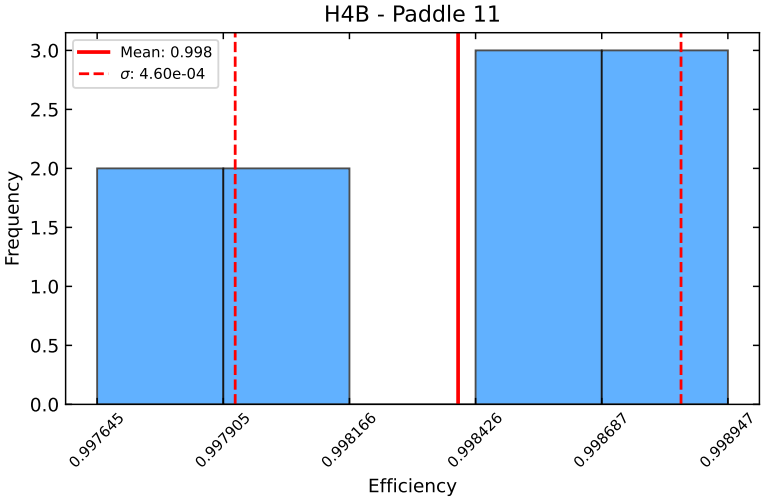
Efficiency Distribution: H4B - Paddle 9



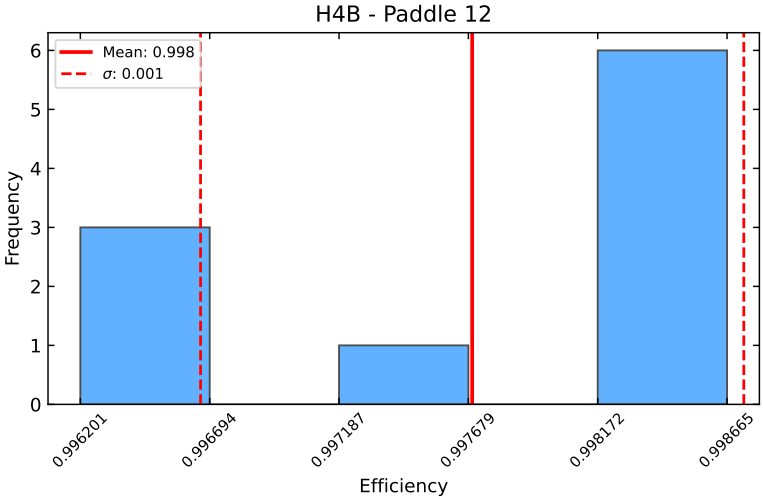
Efficiency Distribution: H4B - Paddle 10



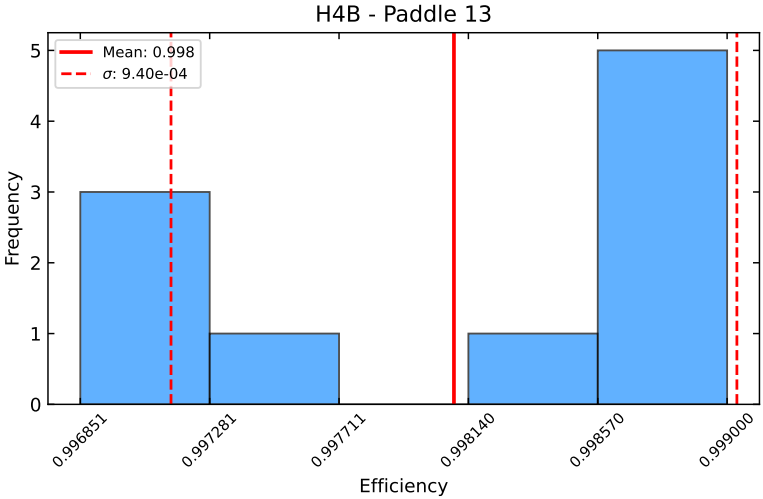
Efficiency Distribution: H4B - Paddle 11



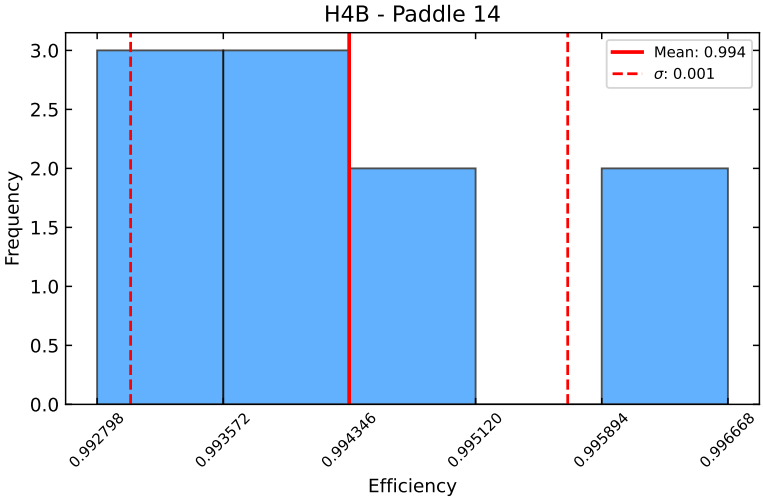
Efficiency Distribution: H4B - Paddle 12



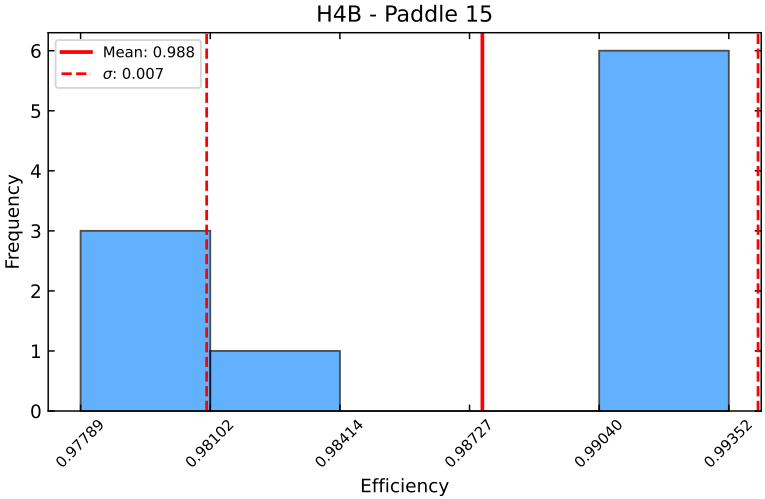
Efficiency Distribution: H4B - Paddle 13



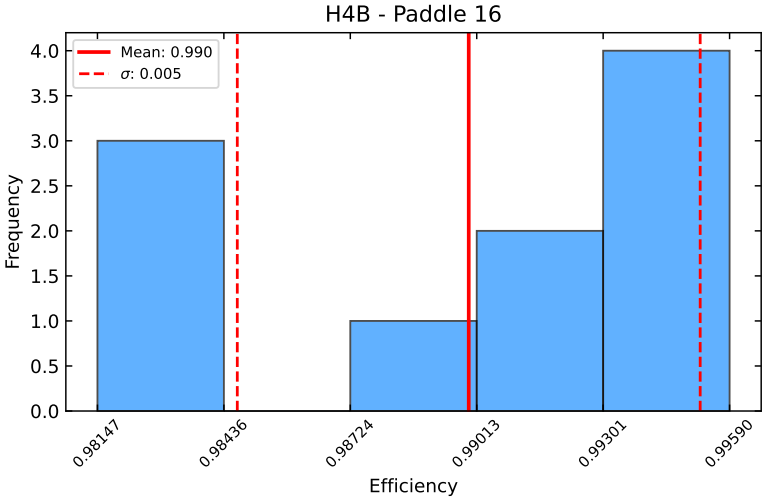
Efficiency Distribution: H4B - Paddle 14



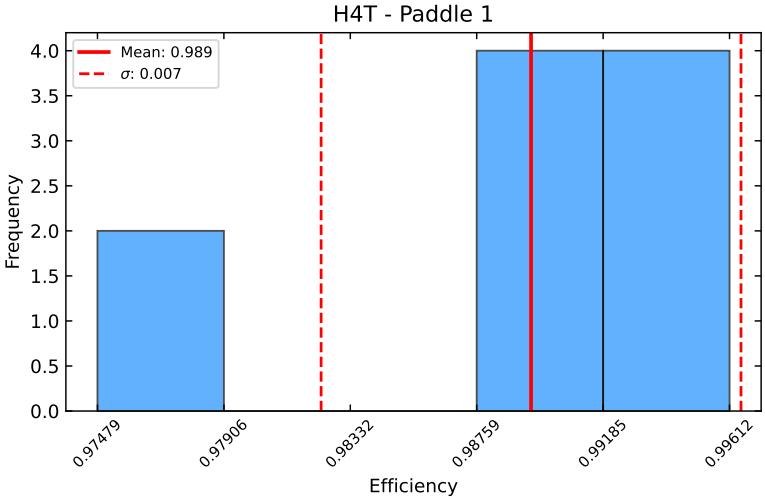
Efficiency Distribution: H4B - Paddle 15



Efficiency Distribution: H4B - Paddle 16



Efficiency Distribution: H4T - Paddle 1



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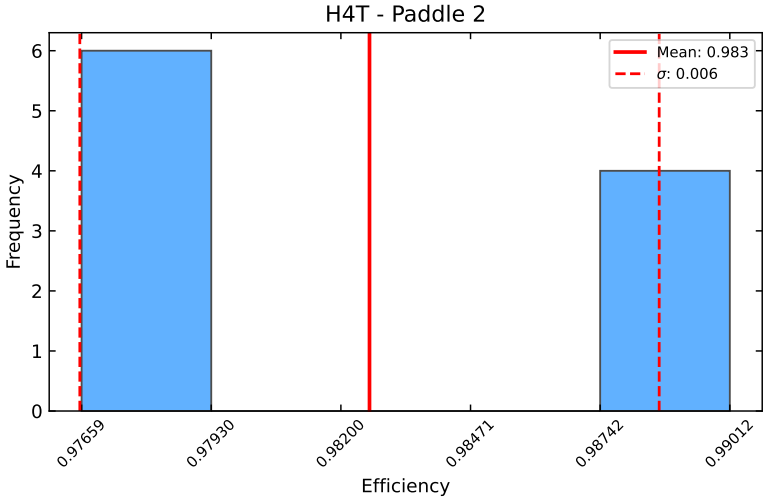
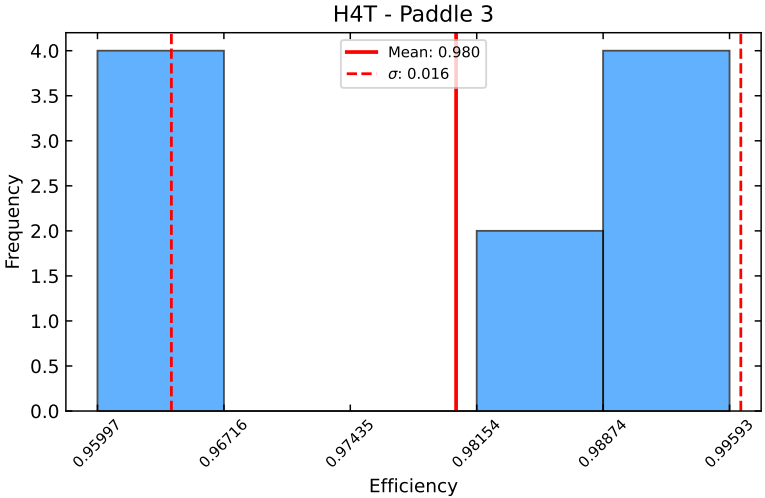
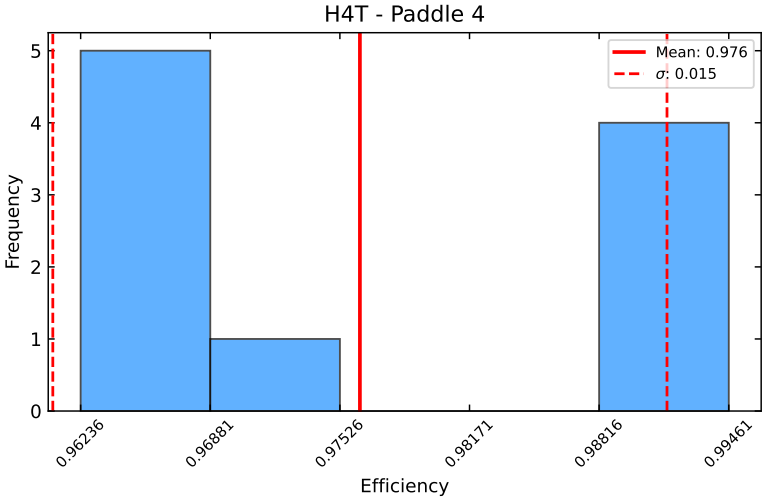


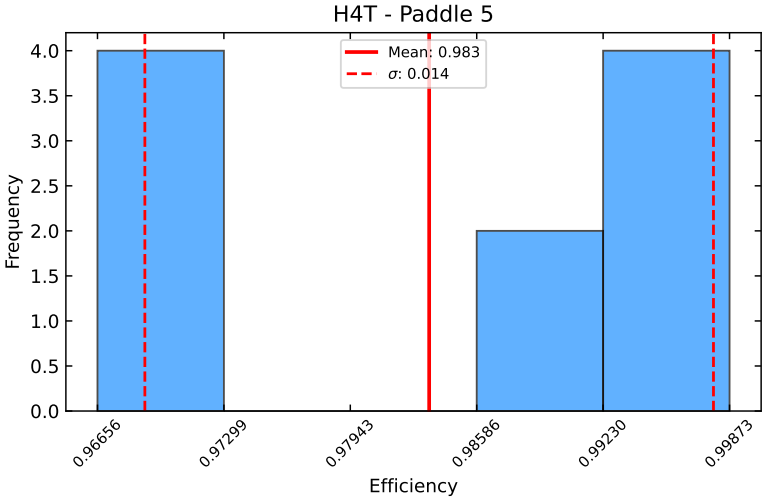
Efficiency Distribution: H4T - Paddle 3



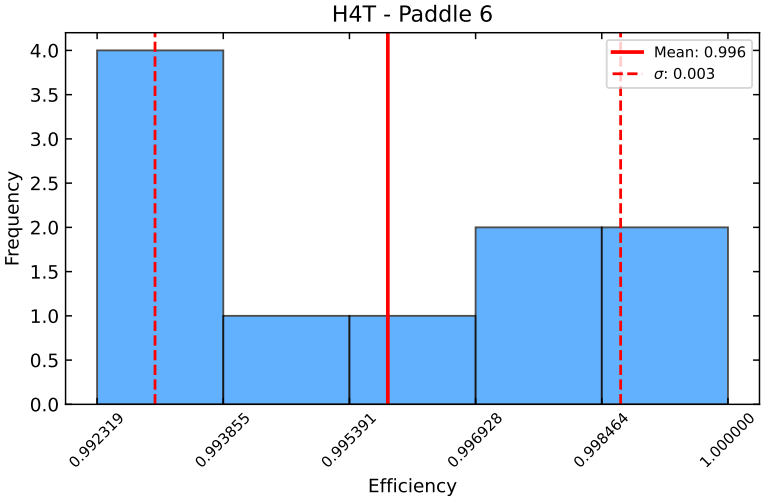
Efficiency Distribution: H4T - Paddle 4



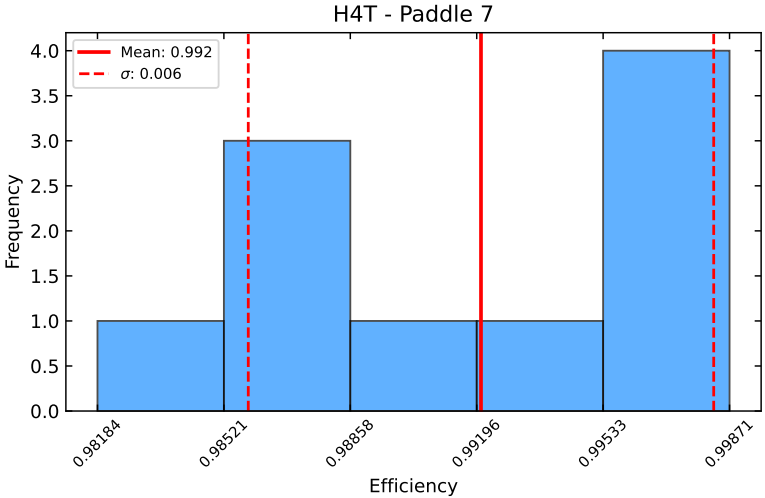
Efficiency Distribution: H4T - Paddle 5



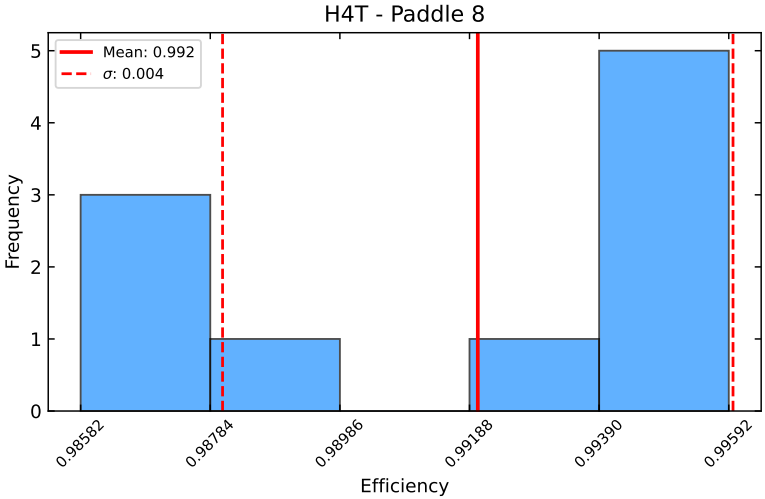
Efficiency Distribution: H4T - Paddle 6



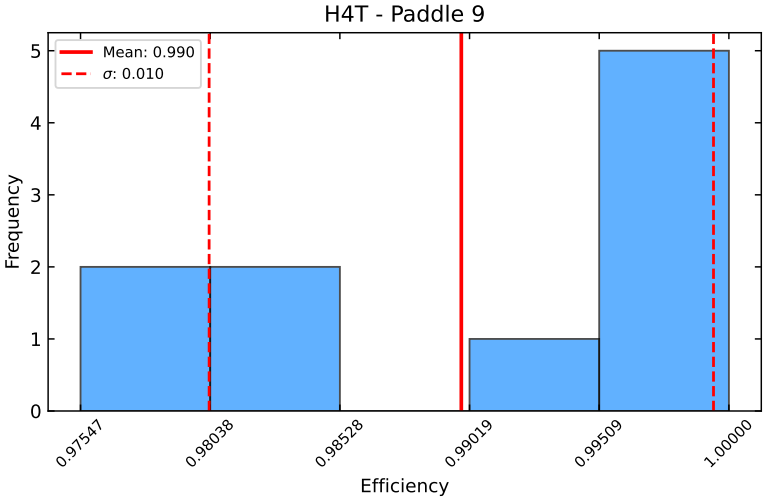
Efficiency Distribution: H4T - Paddle 7



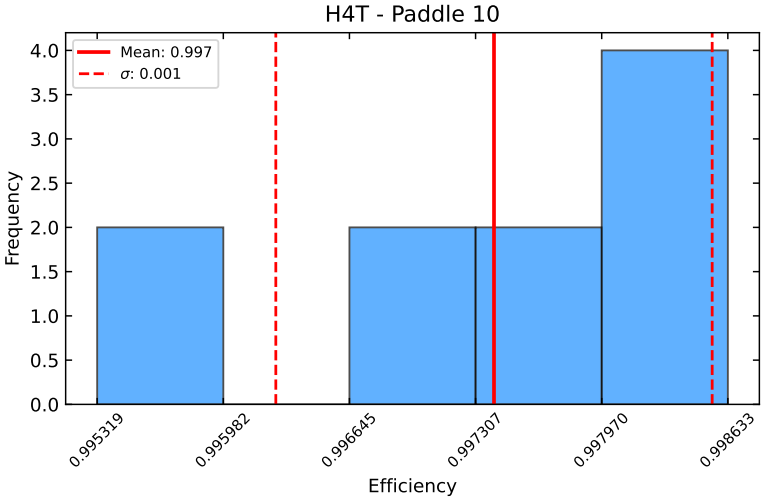
Efficiency Distribution: H4T - Paddle 8



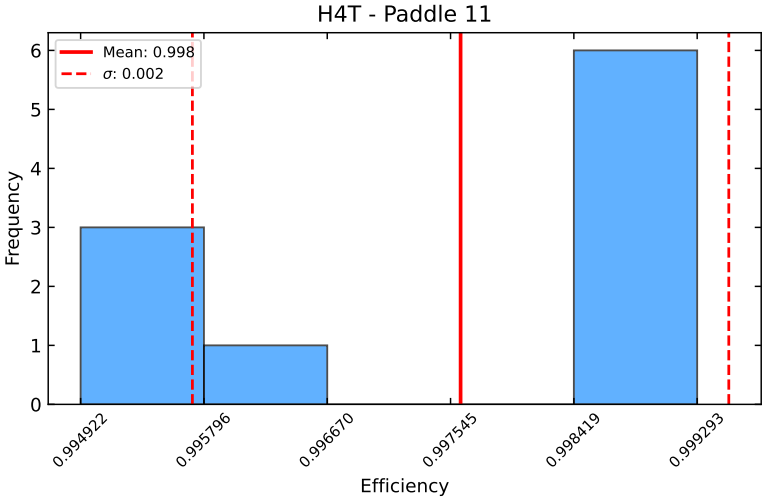
Efficiency Distribution: H4T - Paddle 9



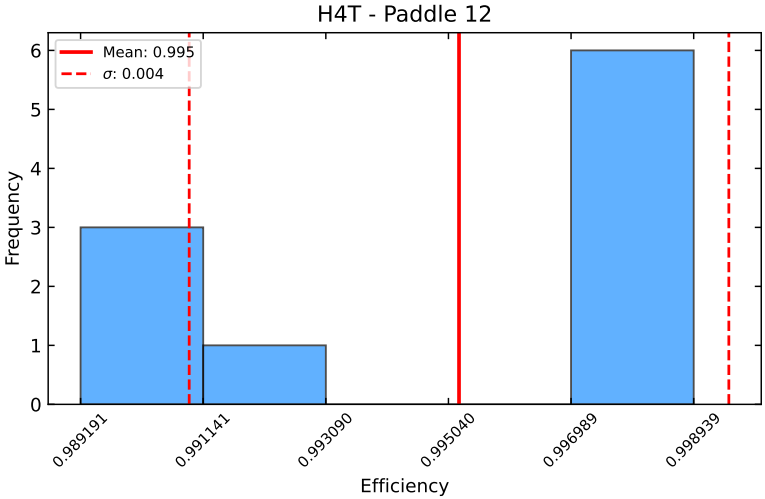
Efficiency Distribution: H4T - Paddle 10



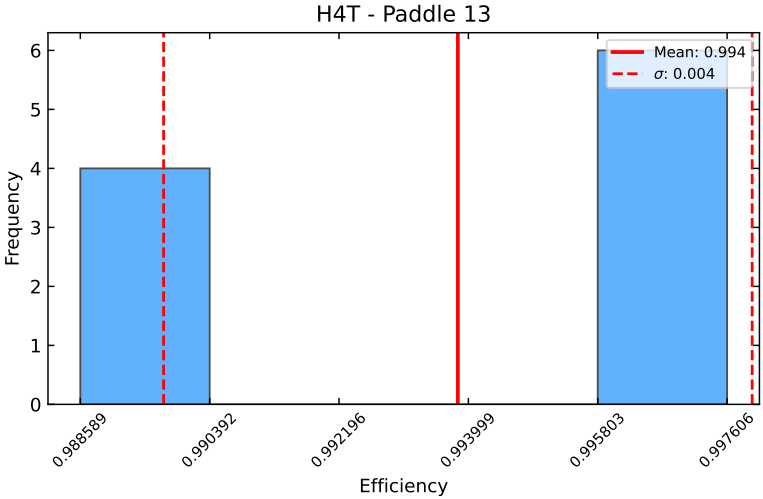
Efficiency Distribution: H4T - Paddle 11



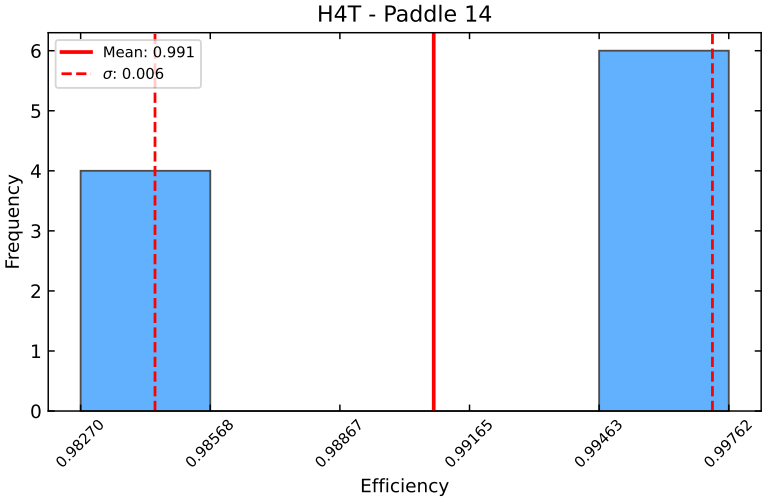
Efficiency Distribution: H4T - Paddle 12



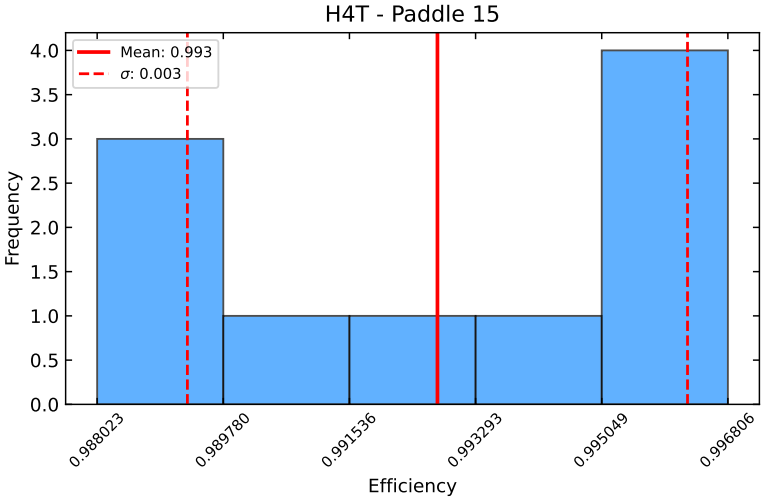
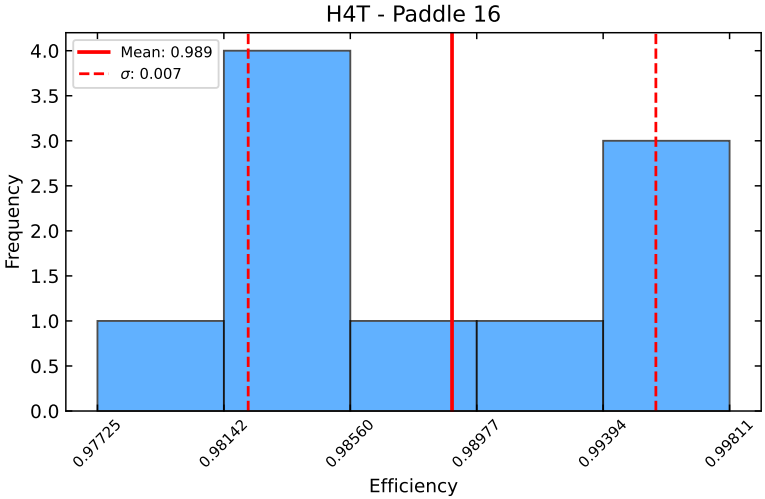
Efficiency Distribution: H4T - Paddle 13



Efficiency Distribution: H4T - Paddle 14



Efficiency Distribution: H4T - Paddle 15



Efficiency Distribution: H4T - Paddle 16